Data Science and Business Analytics

Data Science Case Studies
Lecture 6

Moscow
2025

Optimization problems in Retail. Inventory Optimization Problem. Price Optimization Problem

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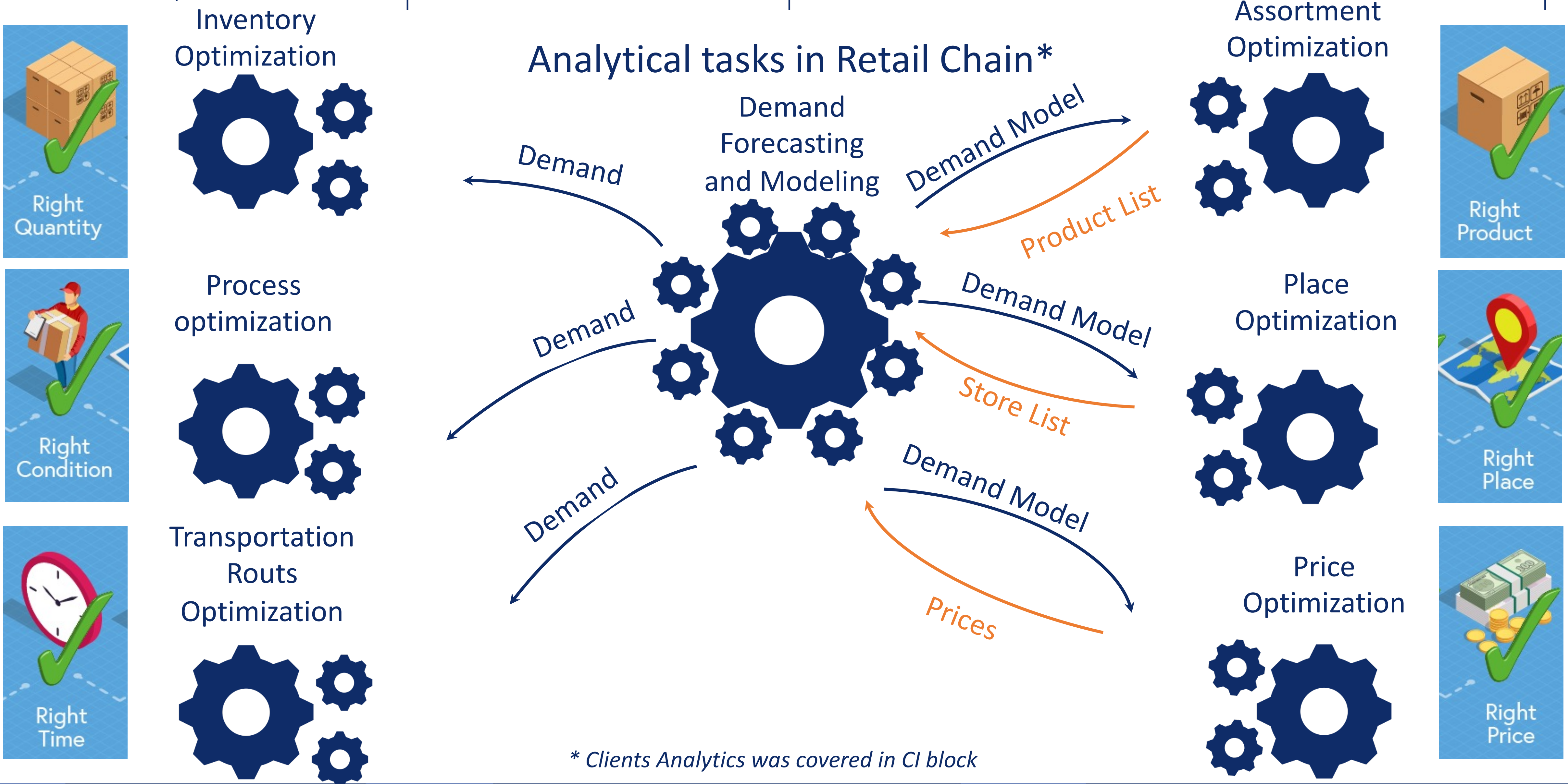
Agenda

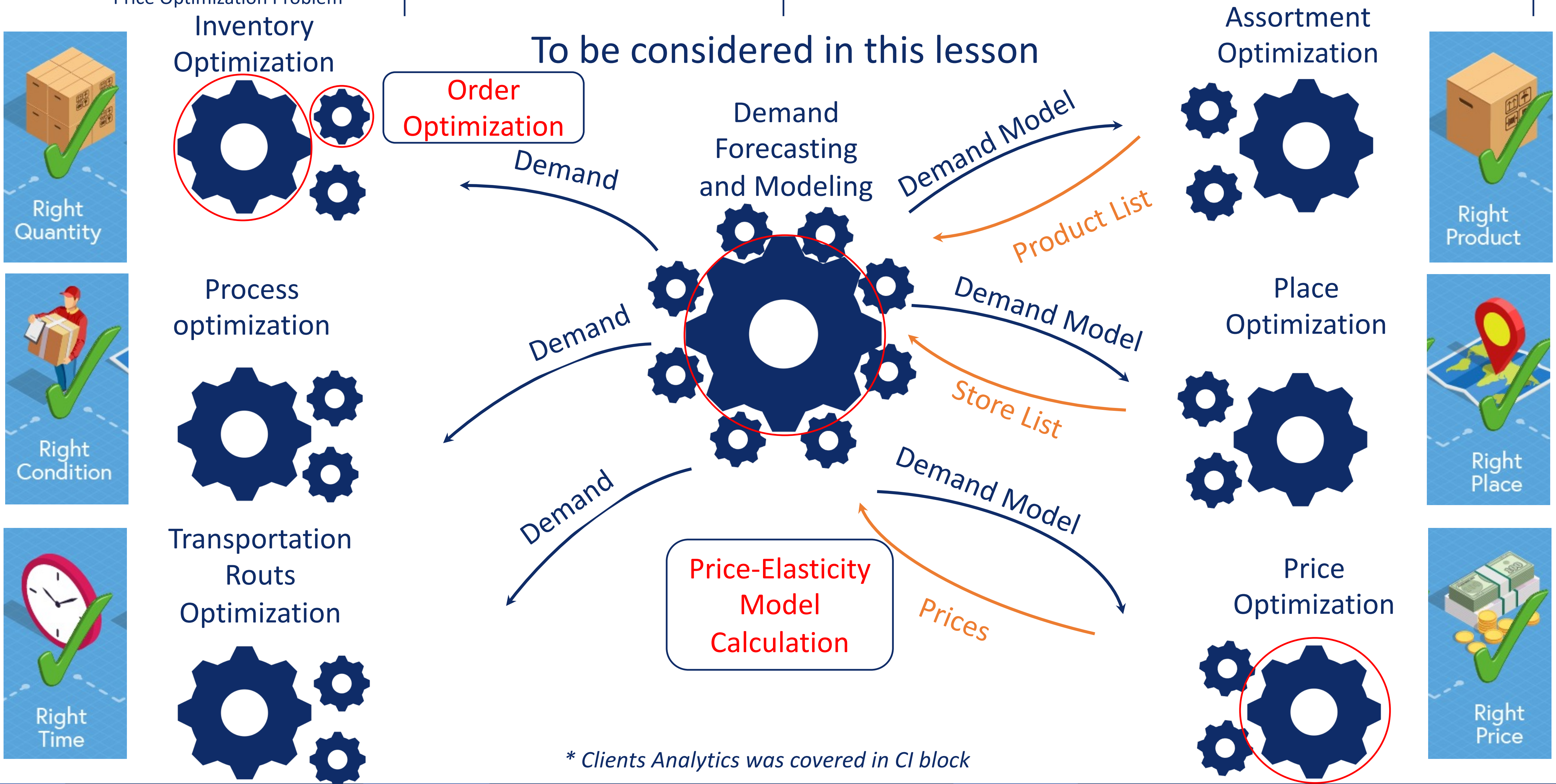
- Price Optimization Problem in Retail
  - Goals of Optimization
  - PED model
  - Solving of Optimization Problem
- Inventory Optimization Problem in Retail
  - Optimization Policy
  - Considering Business Constraints

Analytical tasks in Retail Chain*



** Clients Analytics was covered in CI block*

To be considered in this lesson



** Clients Analytics was covered in CI block*

KPIs of Price Optimization Problem

There are several KPIs:

Revenue = Sales x Price

Gross Profit = Net Revenue – Cost

$$\text{Gross Profit Margin} = \frac{\text{Gross Profit}}{\text{Revenue}}$$

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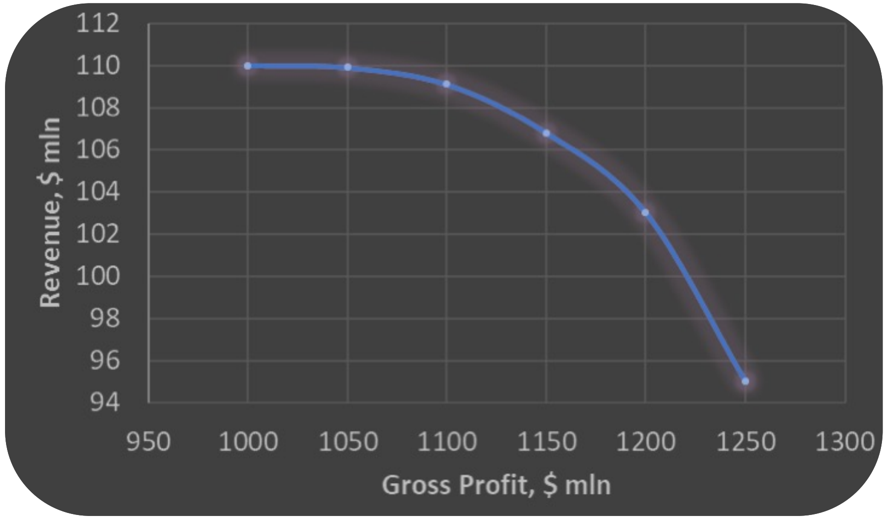
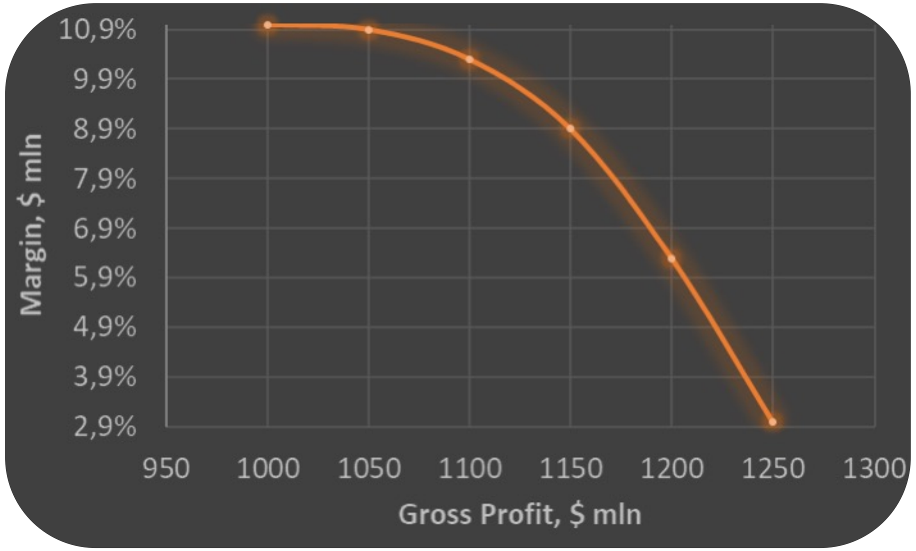
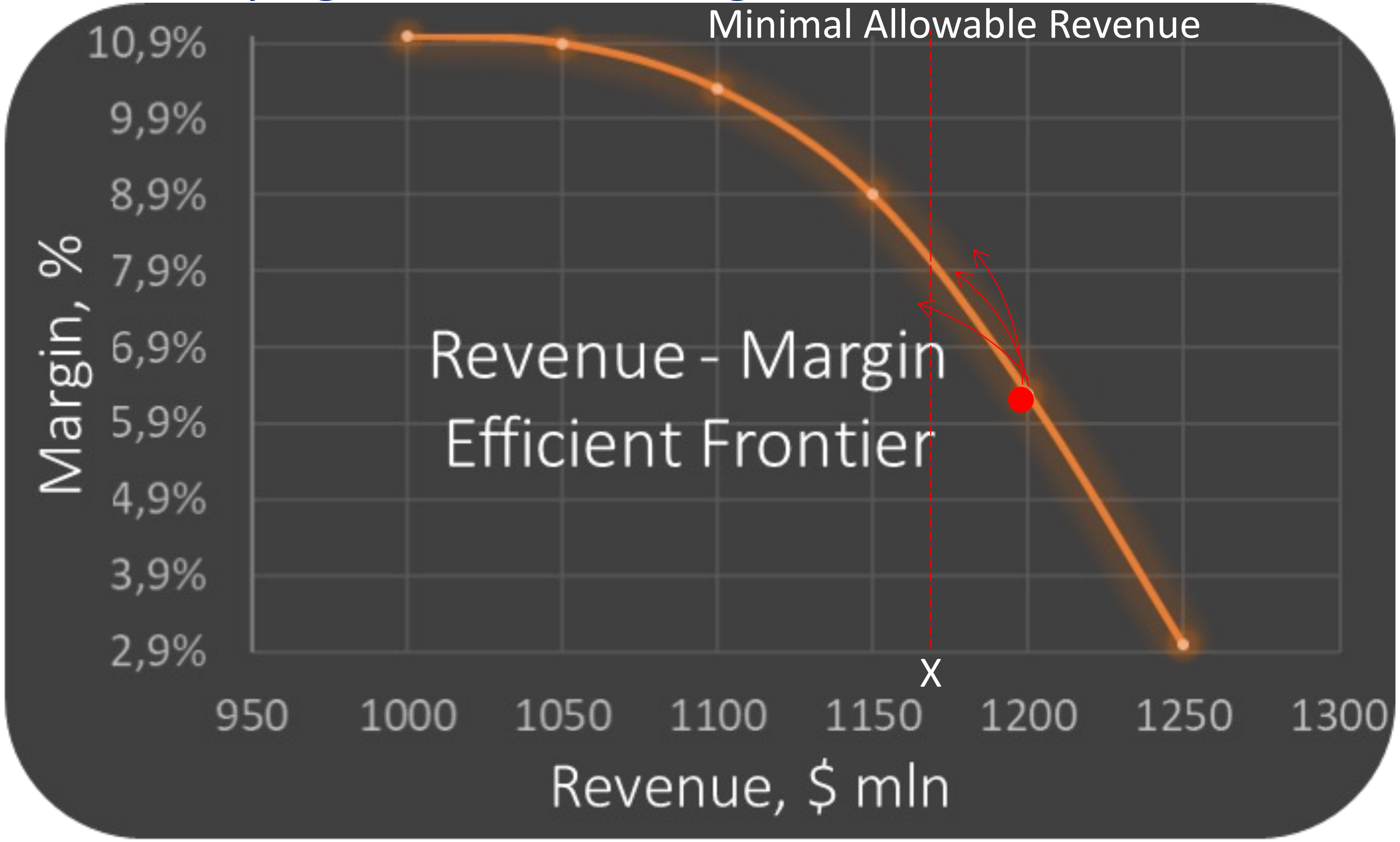




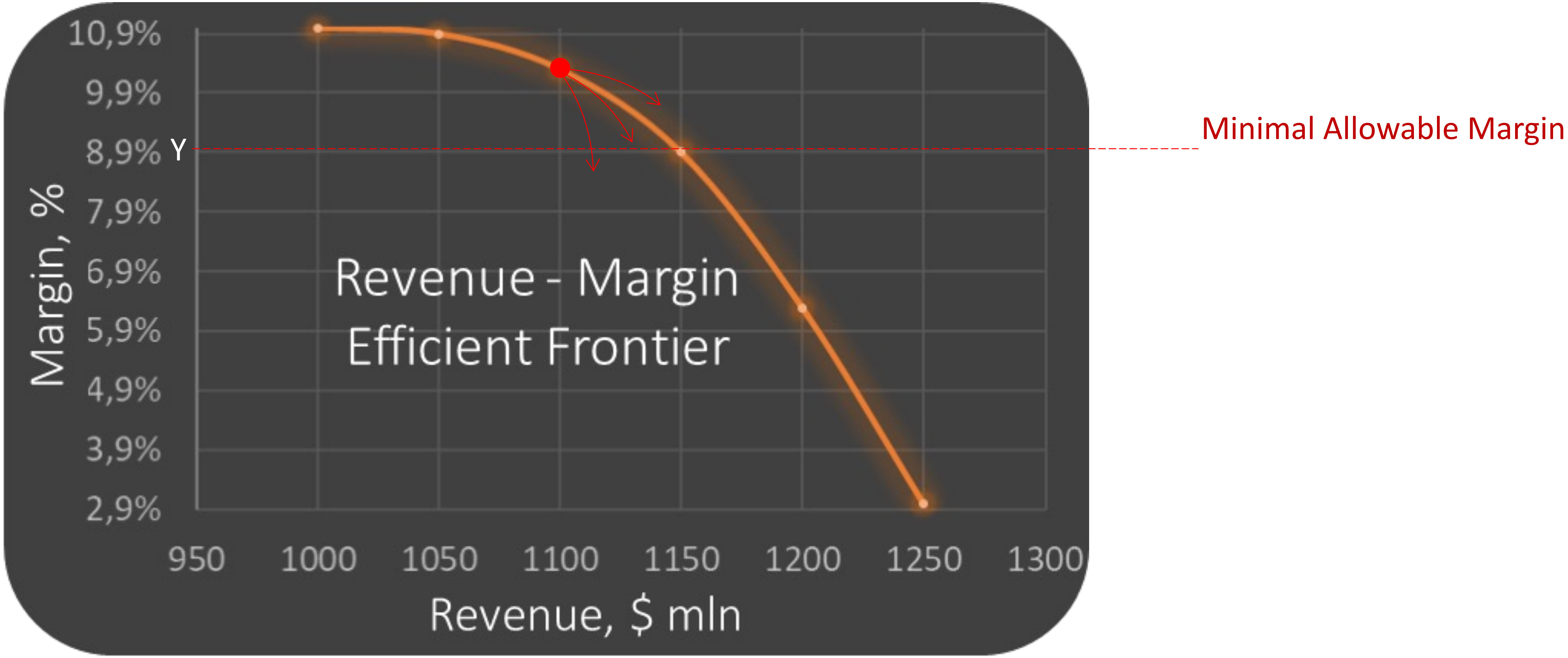
KPIs of Price Optimization Problem
Case 1: Trying to Increase Margin



Constraint: Revenue must be more than X $ mln

KPIs of Price Optimization Problem
Case 2: Trying to Increase Revenue

Constraint: Margin must be more than Y %



Real Optimization Problem

$$Revenue = \sum_{Product,Store,Time} price_{P,S,T} \cdot Quantity(price, \dots) \to \max_{price_{P,S,T}}$$

Constraints

$$\begin{cases} Margin = \dfrac{Revenue(price_{P,S,T}) - Cost}{Revenue(price_{P,S,T})} \geq Y\% \\ ? \\ ? \end{cases}$$

MathOptimization Problem

$$Revenue = \sum_{Product, Store, Time} price_{P,S,T} \cdot Quantity(price, \ldots) \to \max_{price_{P,S,T}}$$

Constraints

$$\begin{cases} Margin = \dfrac{Revenue(price_{P,S,T}) - Cost}{Revenue(price_{P,S,T})} \geq Y\% \\ ? \\ ? \end{cases}$$

Price Elasticity Demand Model

- Quantity = f(price, …)

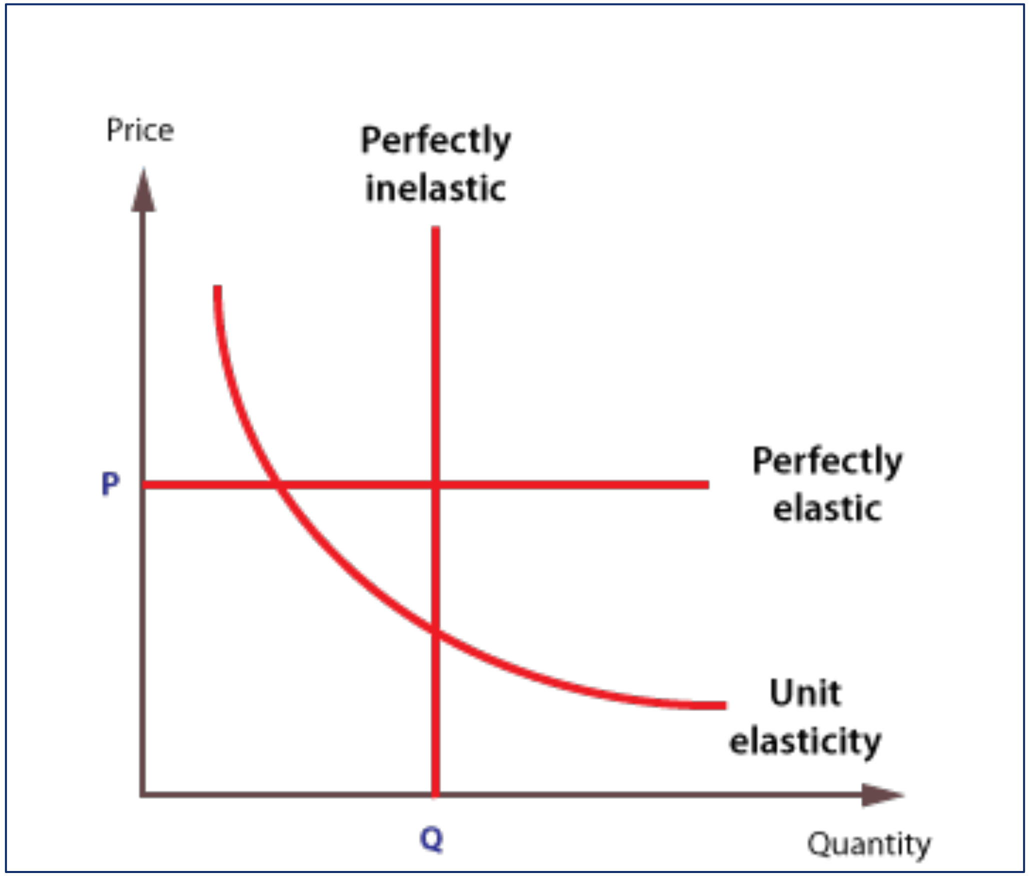
Lin-Lin Price Elasticity Demand model

$$Q = kp + const$$

$$\varepsilon = \frac{\delta Q/Q}{\delta p/p} = \frac{\delta Q}{\delta p} \cdot \frac{p}{Q} = k \cdot \frac{p}{Q}$$

Elasticity Estimation:

$$\bar{\varepsilon} = k * \bar{p}/\bar{Q}$$



Price Elasticity Demand Model

- Quantity = f(price, …)

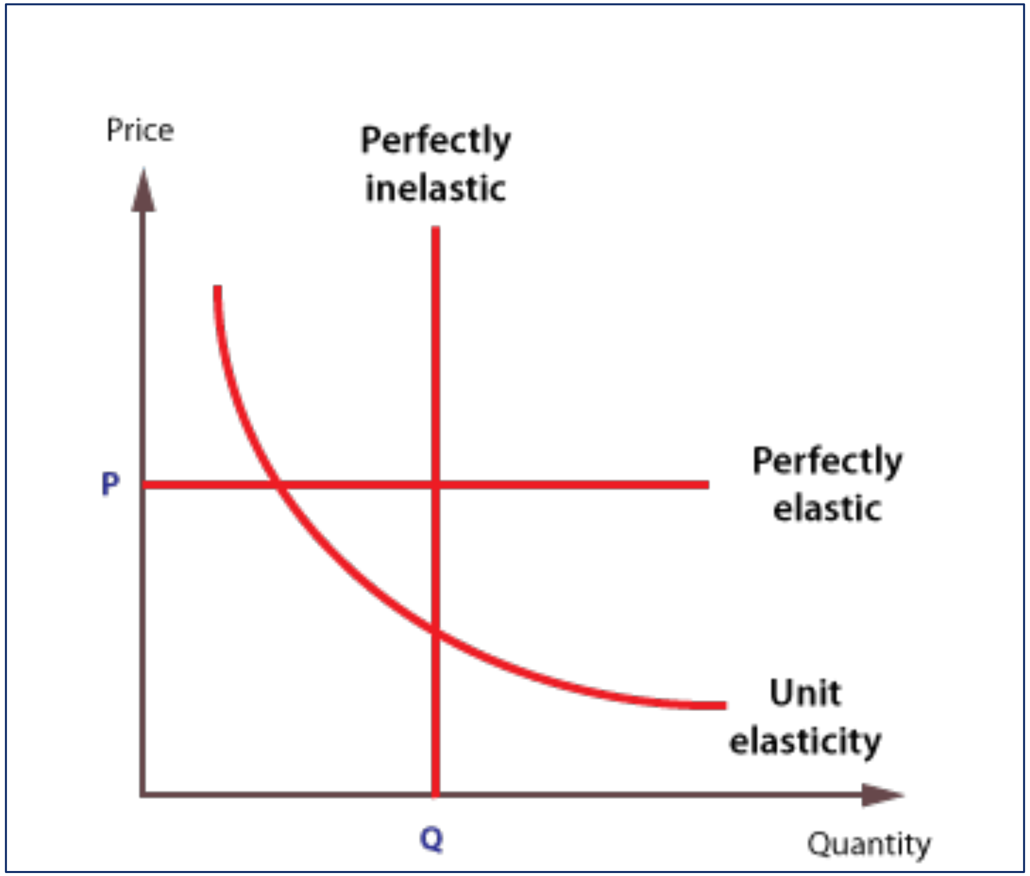
Log-Lin-Lin Price Elasticity Demand model

$$\ln Q = kp + const$$

$$\varepsilon = \frac{\delta Q/Q}{\delta p/p} = \frac{\delta Q}{\delta p} \cdot \frac{p}{Q} = kQ \cdot \frac{p}{Q}$$

Elasticity Estimation:

$$\bar{\varepsilon} = k * \bar{p}$$



Price Elasticity Demand Model

- Quantity = f(price, …)

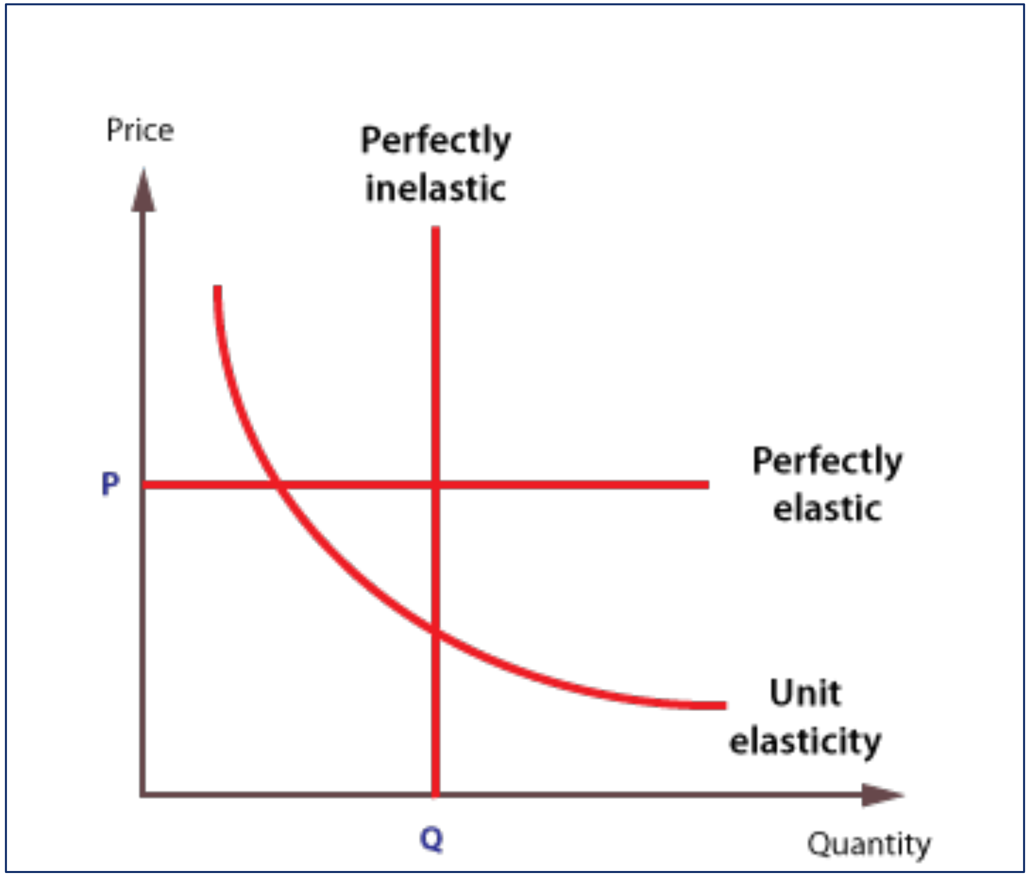
Log-Log Price Elasticity Demand model

$$\ln Q = k \cdot \ln(p) + const$$

$$\varepsilon = \frac{\delta Q/Q}{\delta p/p} = \frac{\delta Q}{\delta \mathrm{p}} \cdot \frac{p}{Q} = \frac{kQ}{p} \cdot \frac{p}{Q}$$

Elasticity Estimation:

$$\bar{\varepsilon} = k$$



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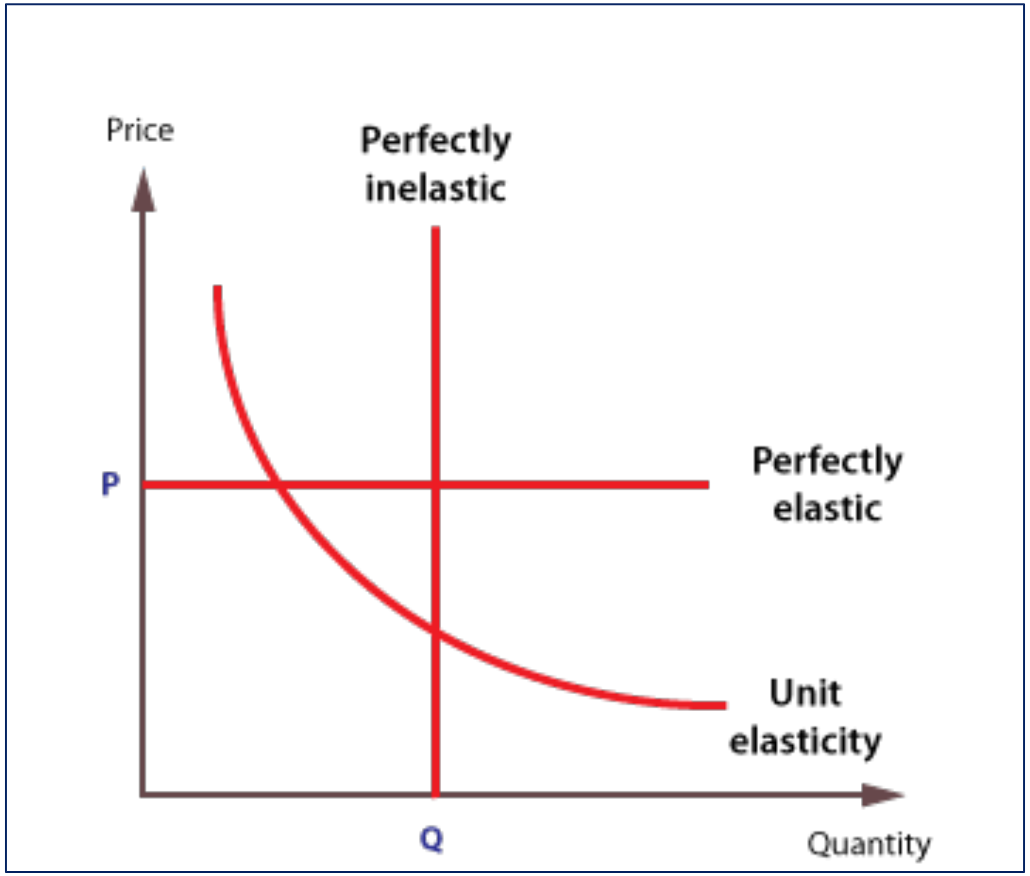
Elasticity Estimation:

$$\bar{\varepsilon} = k$$

Feasible values: $-10 \le \varepsilon \le 0$



Train Set for simple PED model

$$\ln Q = k \cdot \ln(p) + c \cdot 1$$

Store_id	SKU_id	Date	$\ln Q$	$\ln p$	$const$
1	1	01.01.2015	22	163.78	1
1	1	02.01.2015	41	163.78	1
1	1	03.01.2015	35	163.78	1
1	1	04.01.2015	72	163.78	1
1	1	05.01.2015	25	163.78	1

Method:

Train Set for simple PED model

$$\ln Q = k \cdot \ln(p) + c \cdot 1$$

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1	1	05.01.2015	25	163.78	1

Method: OLS (ordinary square regression)

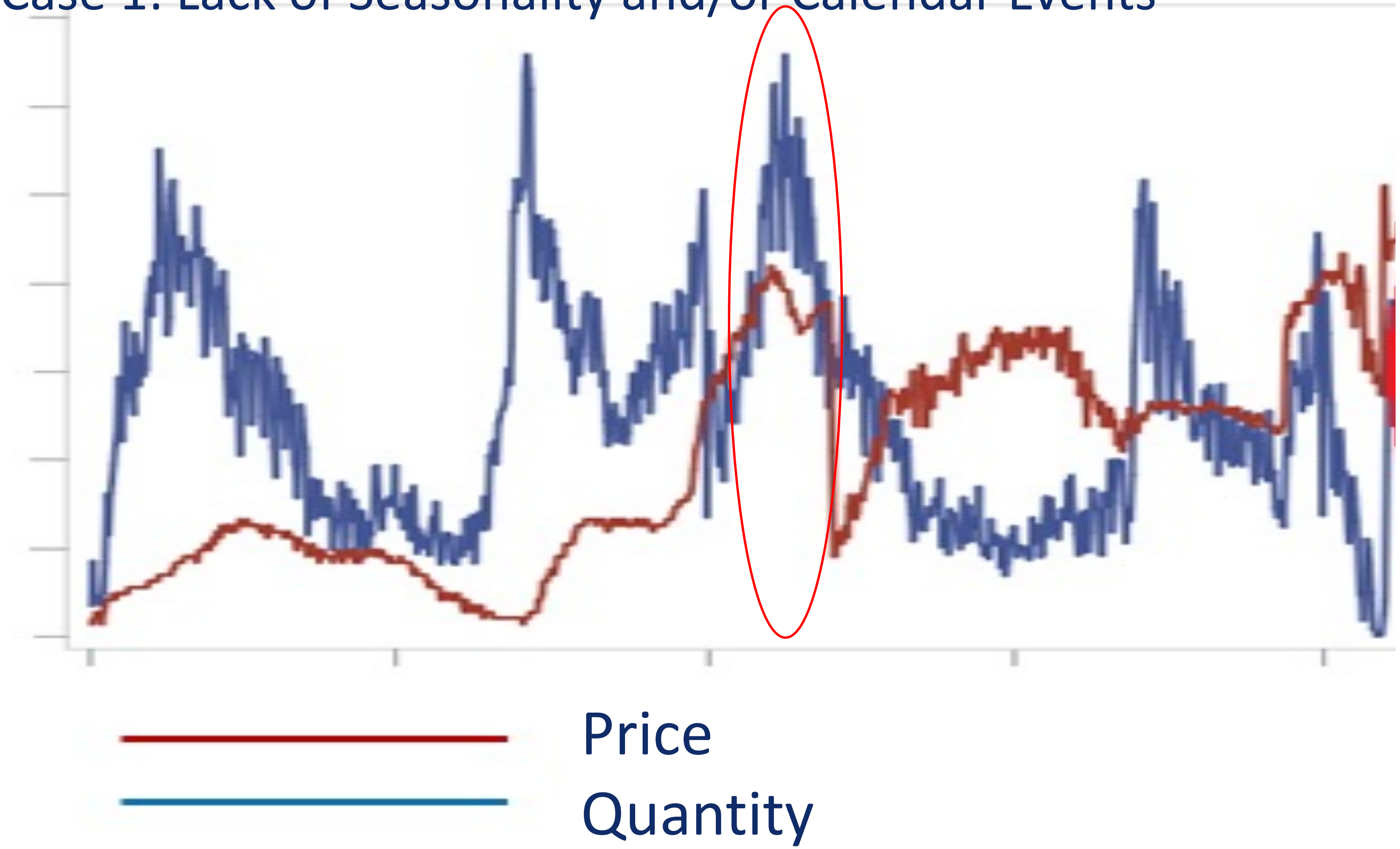
The key issues of classic PED model training

$$\ln Q = k \cdot \ln(p) + c \cdot 1$$

Method: OLS (ordinary least square regression)

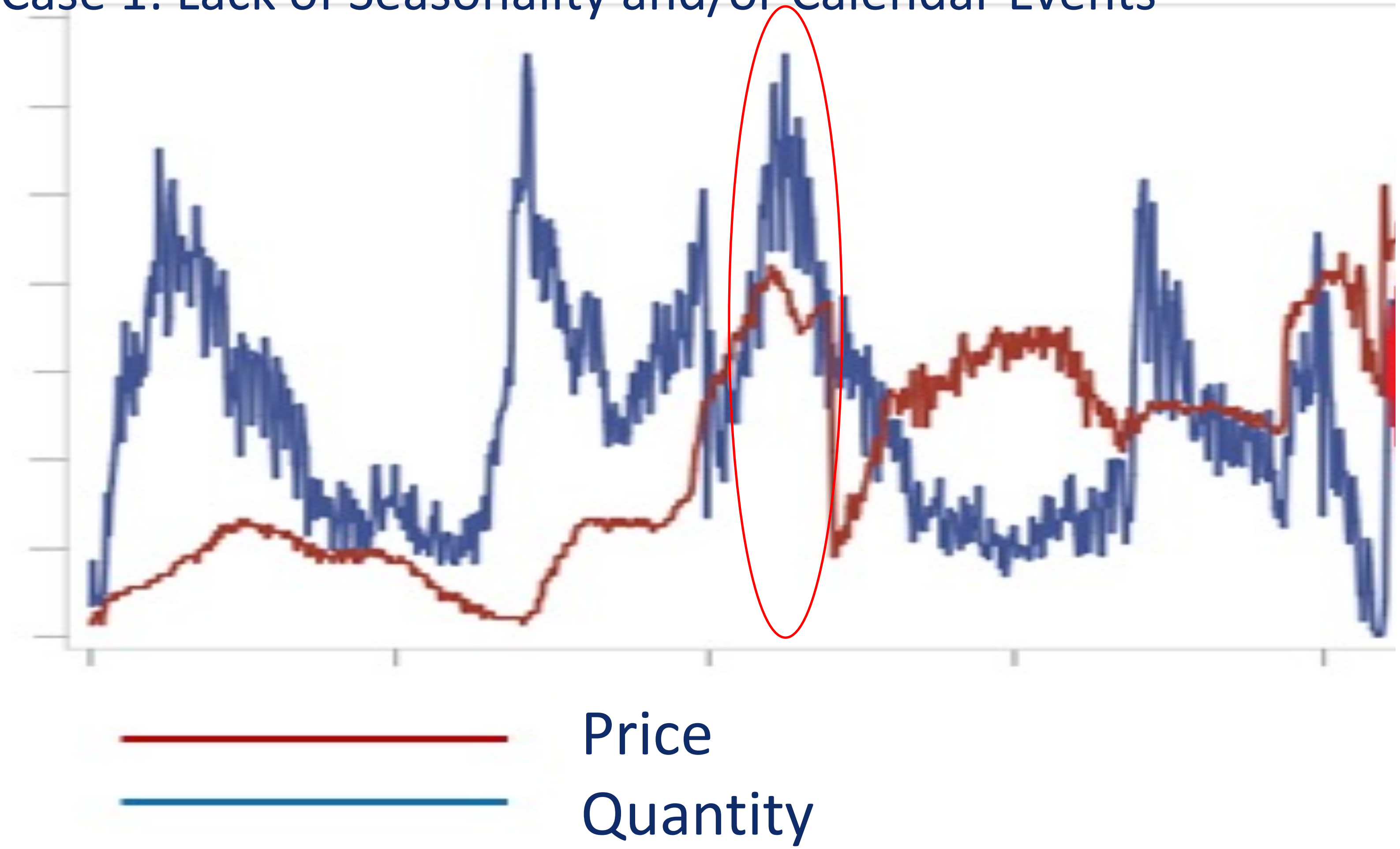
	The issue	Reasons	Solution
1	OLS can not train regression weights (ε *is NaN*)	Small (weak) train set	• Inherit elasticity from similar products (pooling method)
2	Elasticity coefficient is not feasible ($\varepsilon > 0$ o $\varepsilon < -10$))	Lack of features	• Add some features to train set
3	OLS gives a poor quality of regression ($-10 \le \varepsilon \le 0$) but p_value > 0.05	Incorrect grouping products and locations to train set	• Increase train set (add samples from other products/locations) • Exclude inappropriate periods/samples from train set

Price Elasticity Demand Model issues
Case 1: Lack of Seasonality and/or Calendar Events



How to consider:

Price Elasticity Demand Model issues
Case 1: Lack of Seasonality and/or Calendar Events

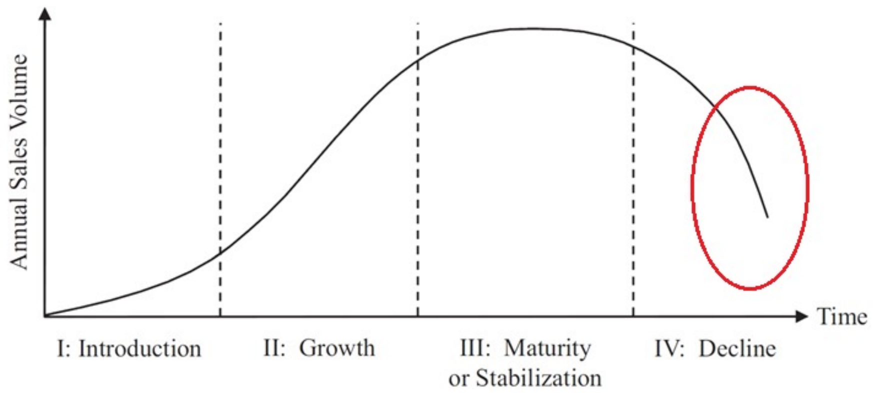


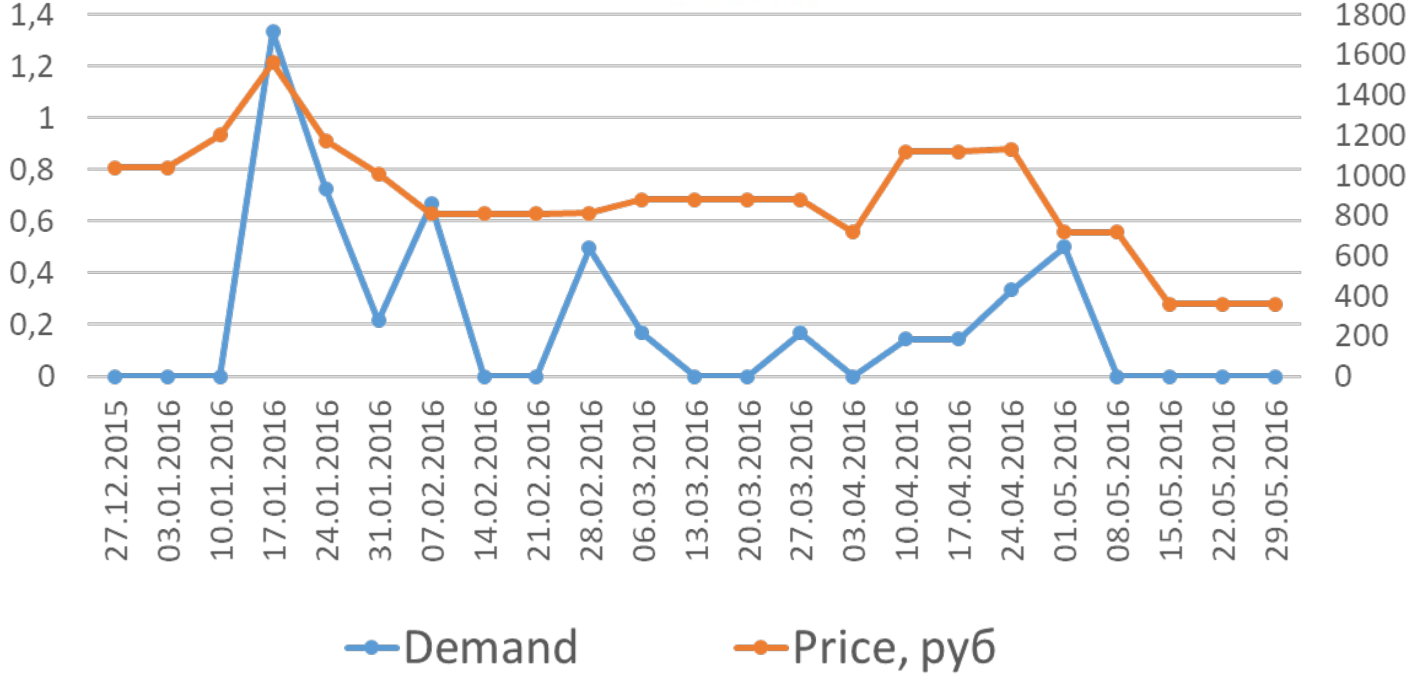
How to consider:

- Data Preprocess: calculate seasonal component (STL decomposition) and extract it from Quantity before PED model
- Add dummy variable to regression

Price Elasticity Demand Model Issues
Case 2: Product LifeCycle Phases

How to consider:

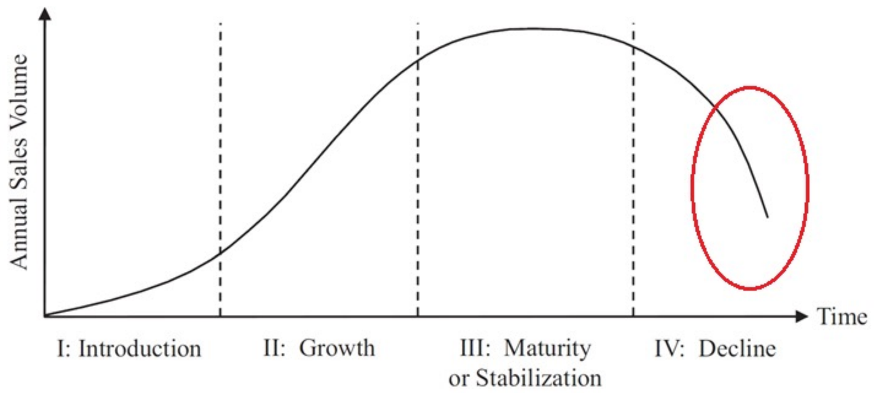


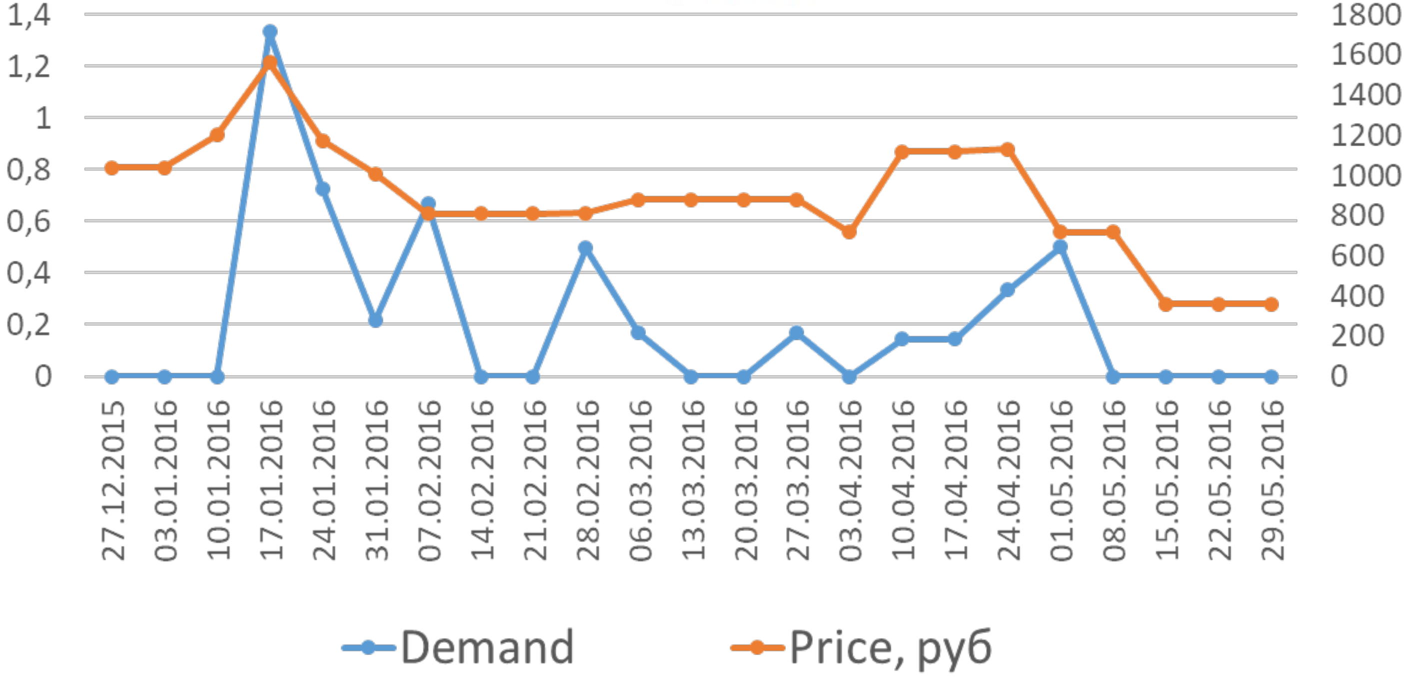


Price Elasticity Demand Model Issues
Case 2: Product LifeCycle Phases

How to consider:

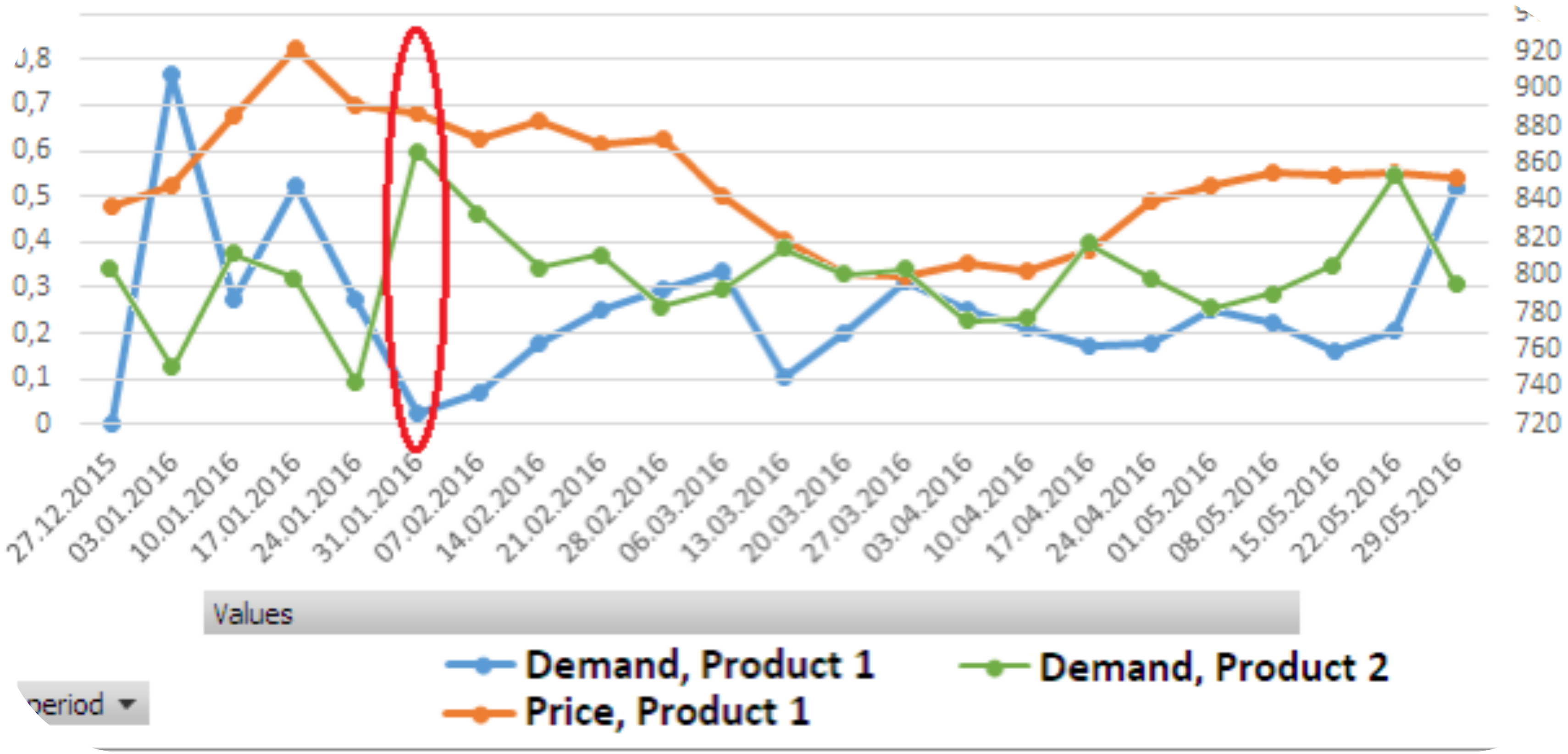
- Filter-out inappropriate phases dates





Price Elasticity Demand Model issues
Case 3: cross cannibalization

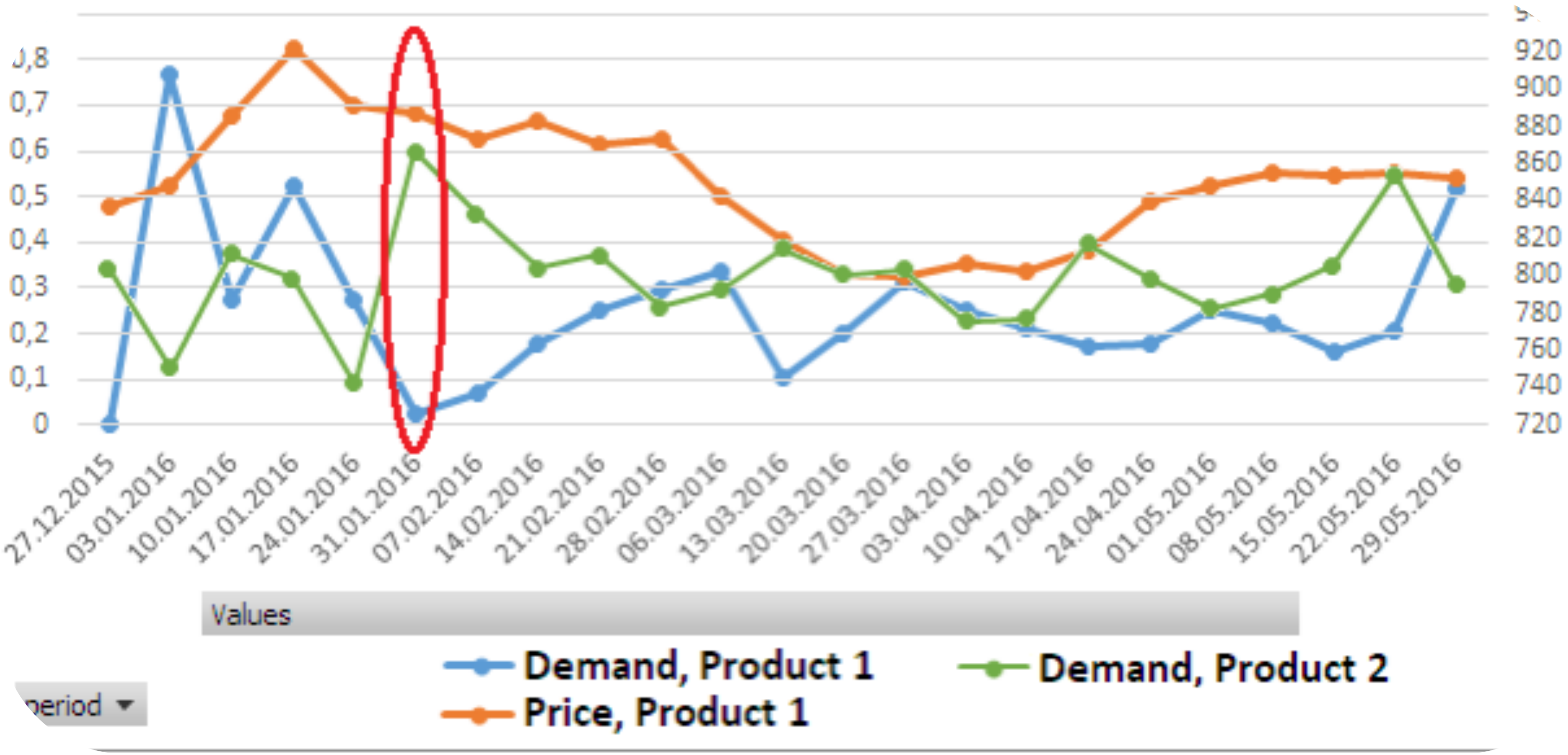
How to consider:



Price Elasticity Demand Model issues
Case 3: cross cannibalization

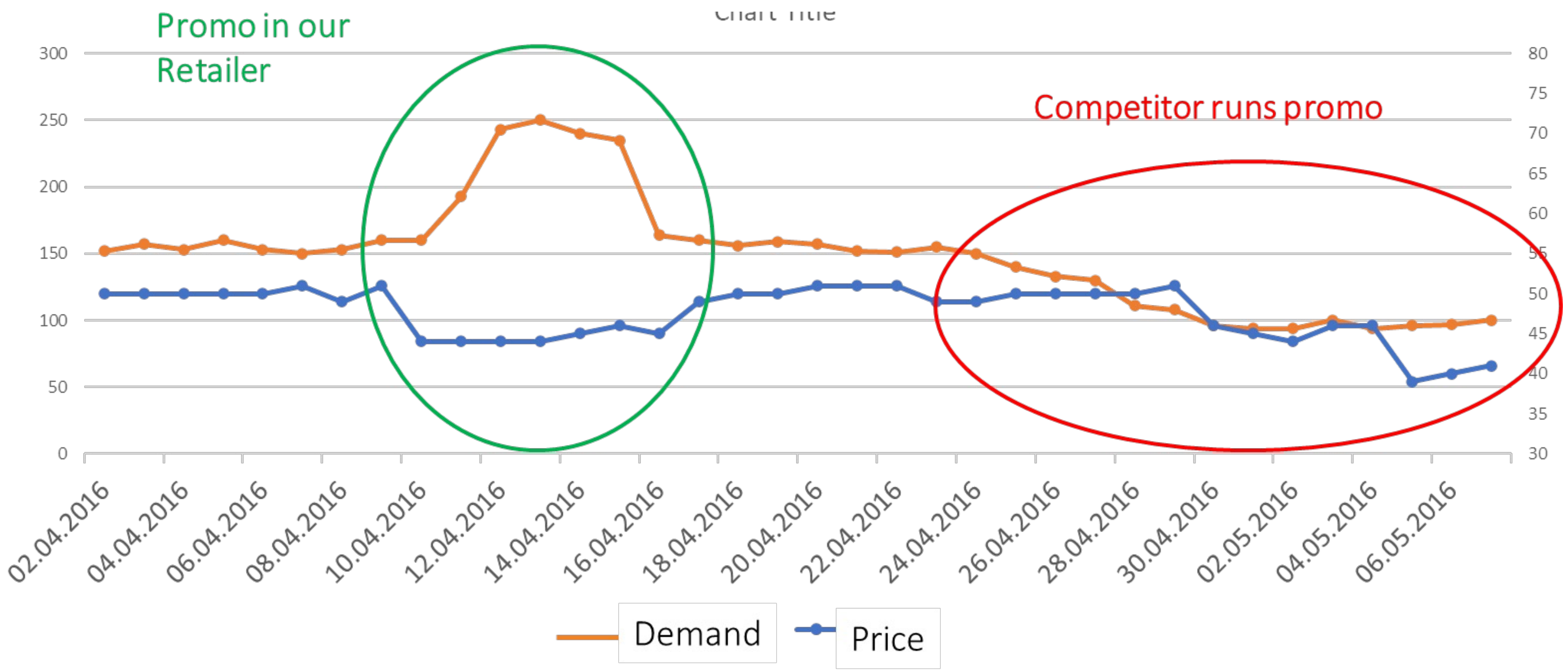
How to consider:

- Add price/promo flag variables related to cannibalization products



Price Elasticity Demand Model Issues
Competitor prices

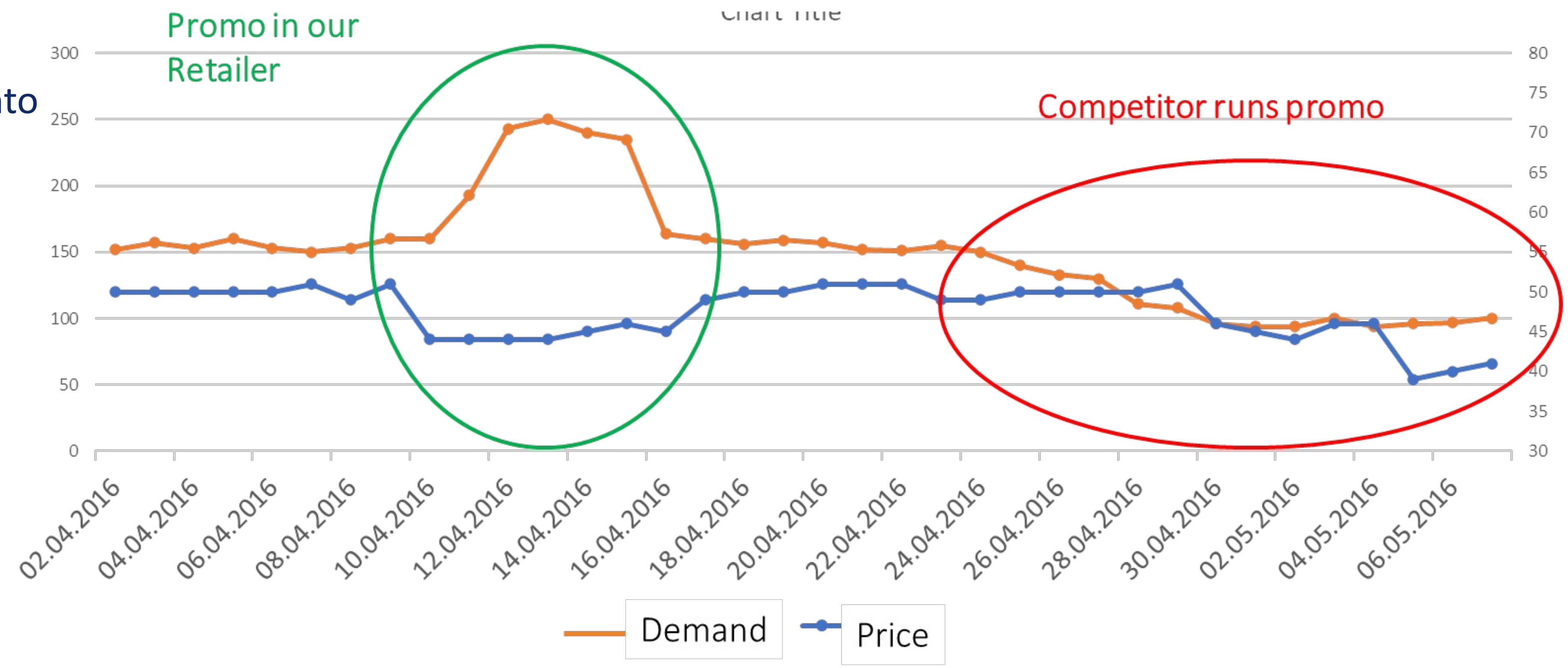
How to consider:



Price Elasticity Demand Model Issues
Competitor prices

How to consider:

- Add competitor-price into PED model
- Provide specific constraints into Optimization Problem



PED model Sum Up

$$Quantity = f\left(Price, Promo, Seasonal, Holiday, Price^{cannib}, \ldots\right)$$

Log–Log model:

$$\log(Sales_t) - Seasonality$$
$$= \alpha + k \cdot \ln(Price_t) + \gamma \cdot Promo_t + \sum \delta \cdot Holiday_t + \sum \zeta \cdot price_t^{cannib} +$$
$$+ \sum \eta \cdot price_t^{competitor} + \cdots$$

Constraints:

$$\begin{cases} k \le 0 \\ \gamma, \zeta, \eta \ge 0 \end{cases}$$

Real Optimization Problem

$$Revenue = \sum_{Product,Store,Time} price_{P,S,T} \cdot \exp\left(\alpha + k \cdot \ln(Price_t) + \sum \zeta \cdot Price_t^{cannib} + \cdots\right)\left(\alpha + k \cdot \ln(Price_t) + \sum \zeta \cdot Price_t^{cannib} + \cdots\right) \to \max_{Price_{P,S,T}}$$

Constraints

$$\begin{cases} Margin = \dfrac{Revenue(price_{P,S,T}) - Cost}{Revenue(price_{P,S,T})} \geq Y\% \\ A \leq price_{P,S,T} \leq B \end{cases}$$

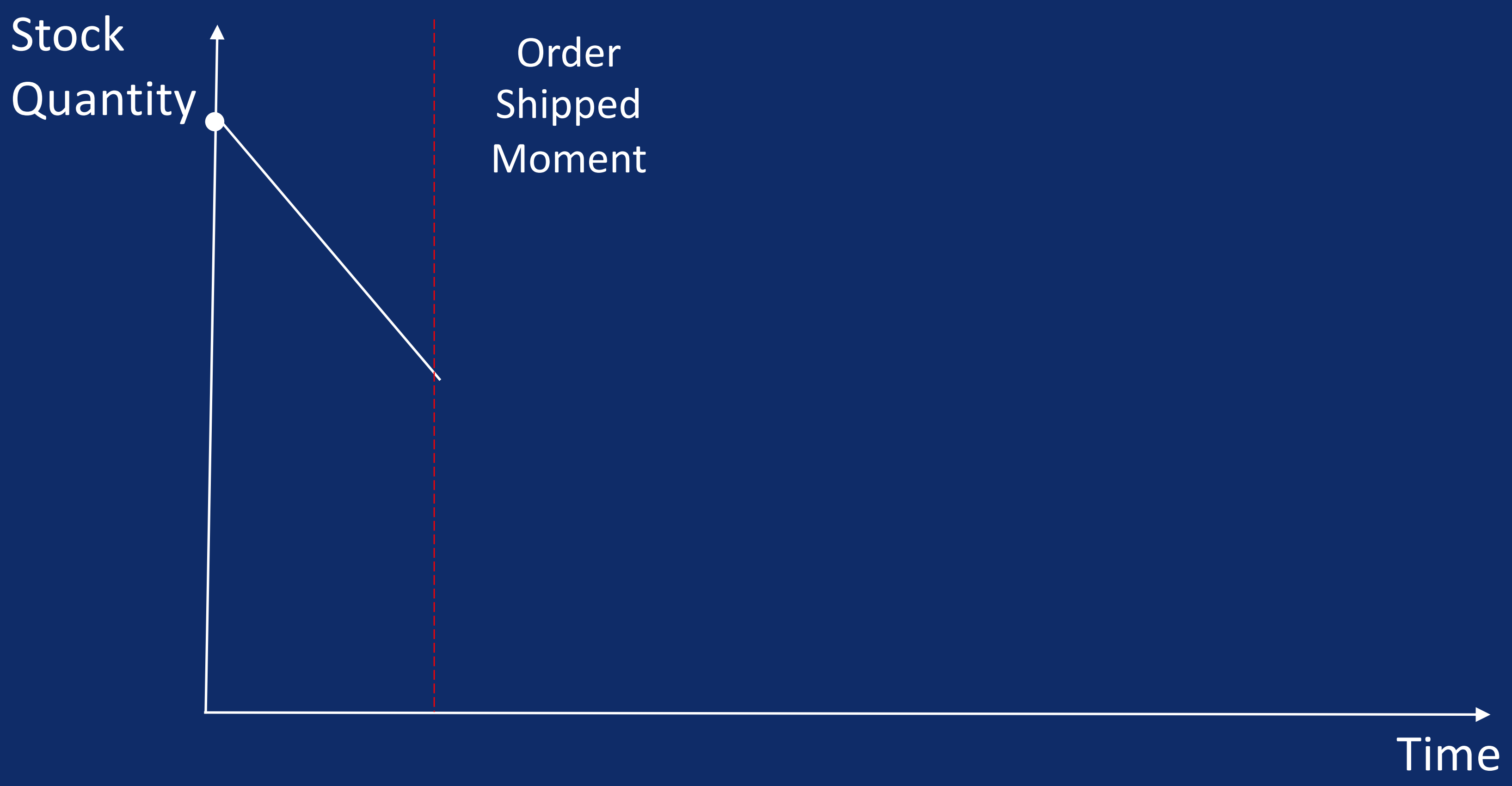
$A, B, \boldsymbol{Y}$ are defined by busuness-user,
A, B are defined based on competitor prices

Agenda

- Inventory Optimization Problem in Retail
    - Optimization Policy
    - Considering Business Constraints

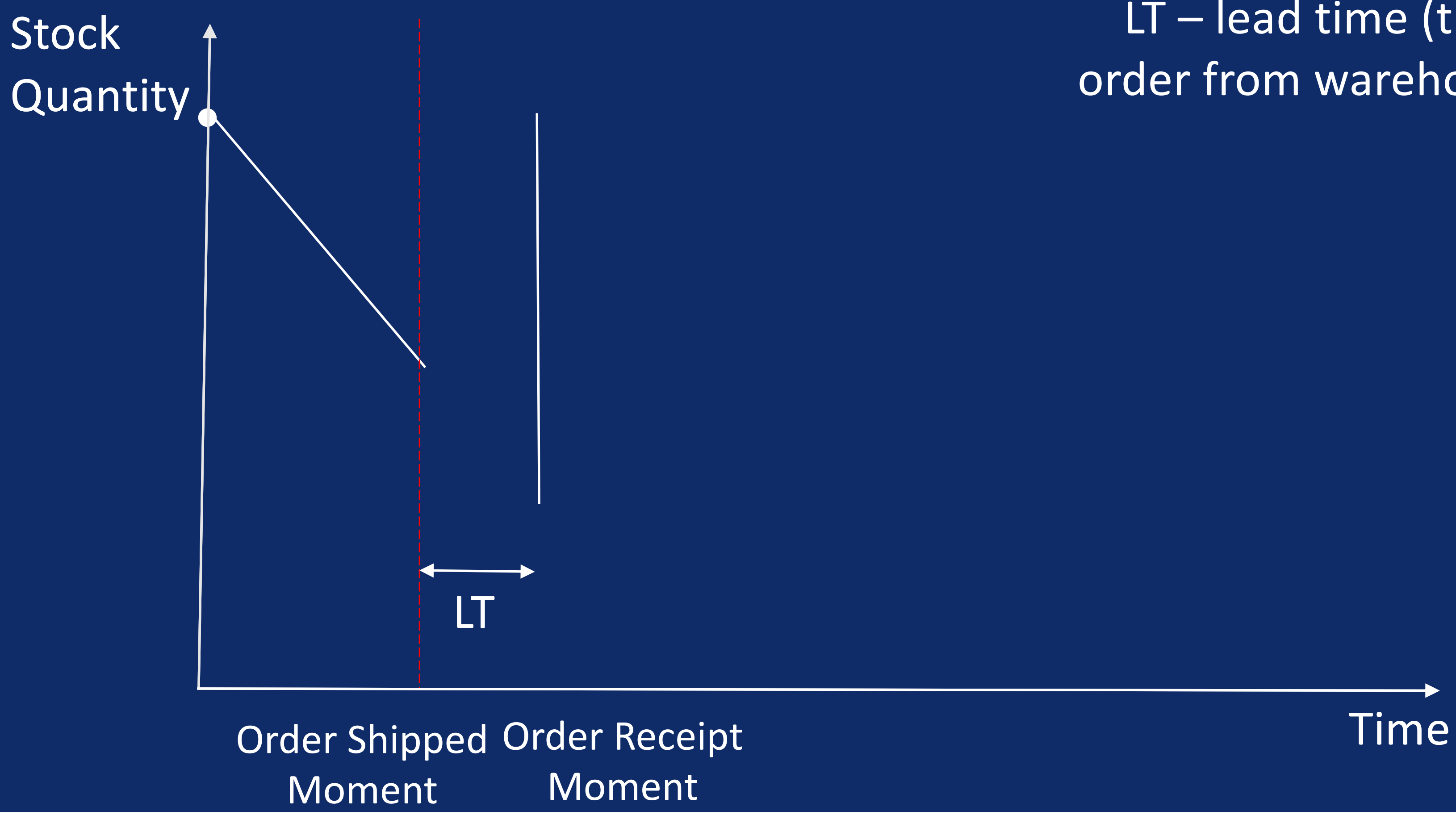
Inventory Management Story

Inventory Management Story



Inventory Management Story

LT – lead time (time to ship order from warehouse to store)



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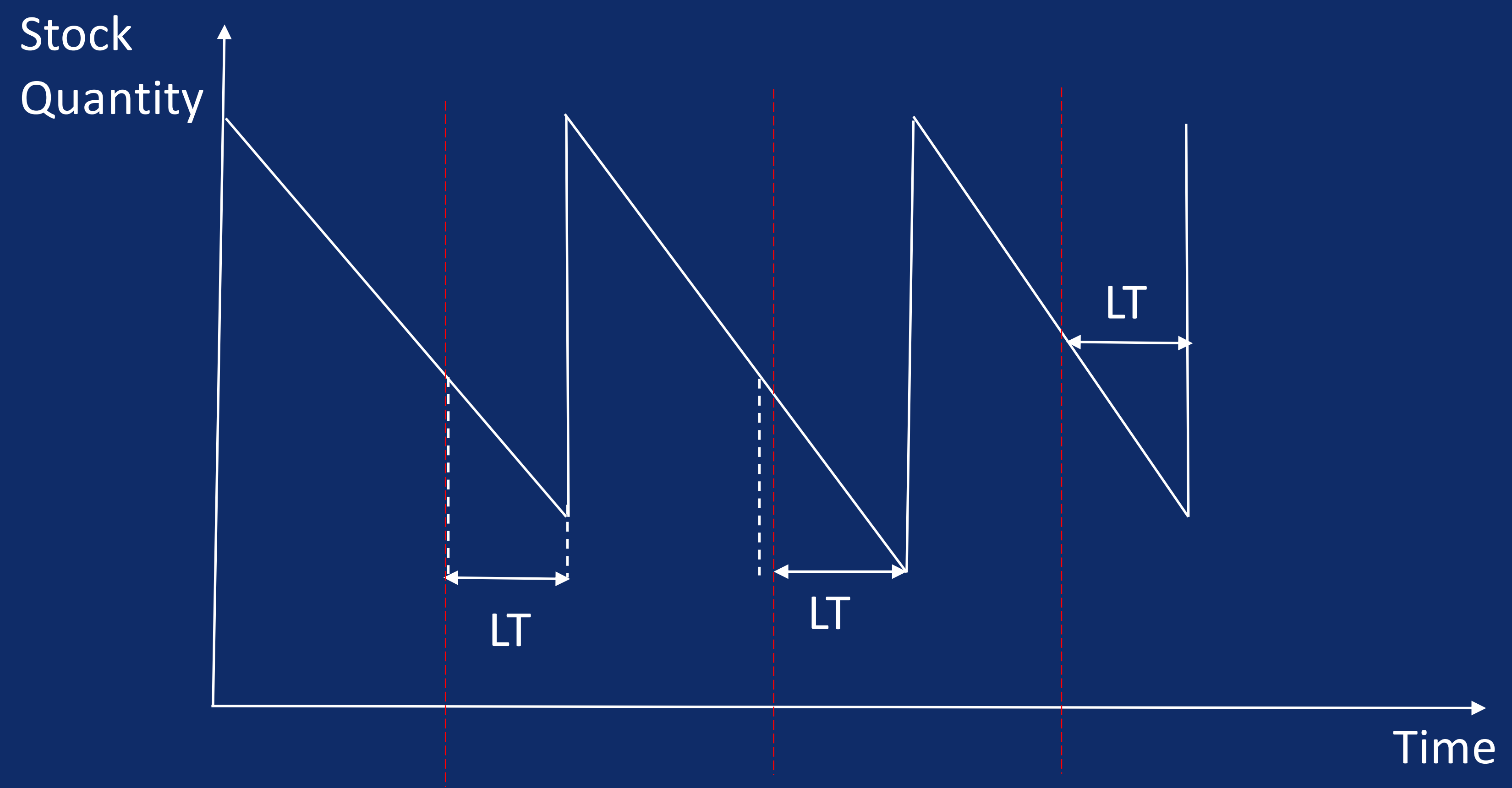

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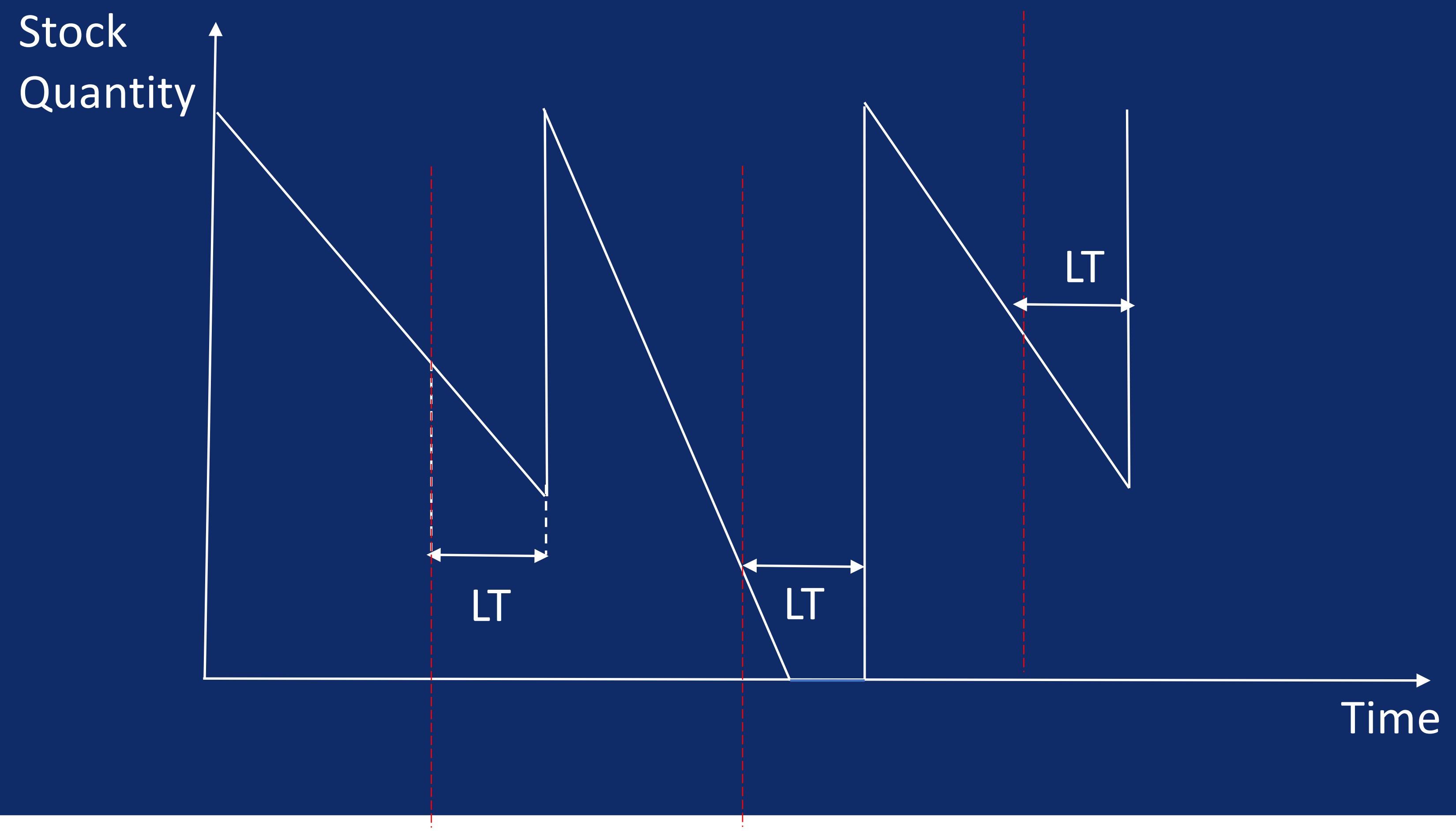

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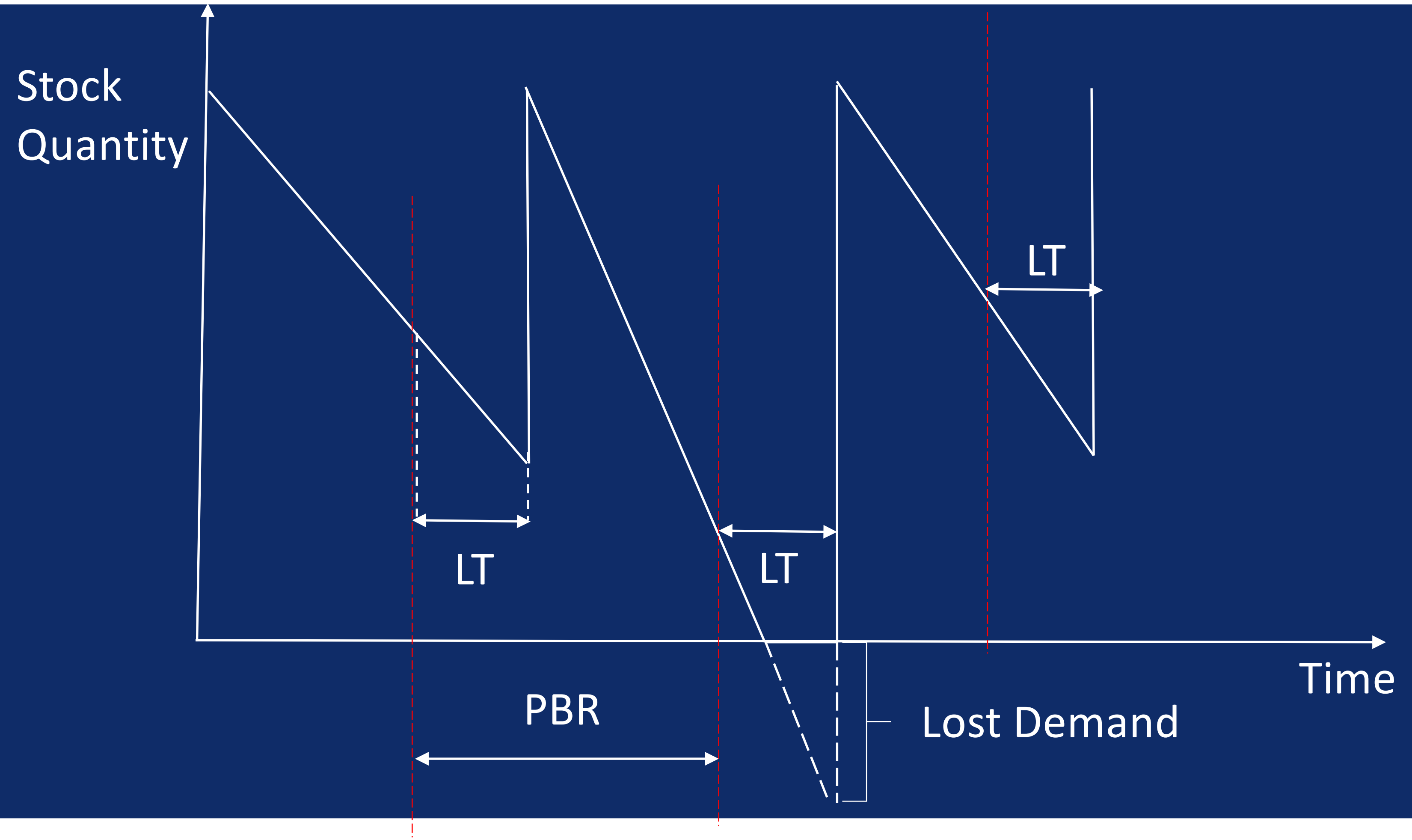
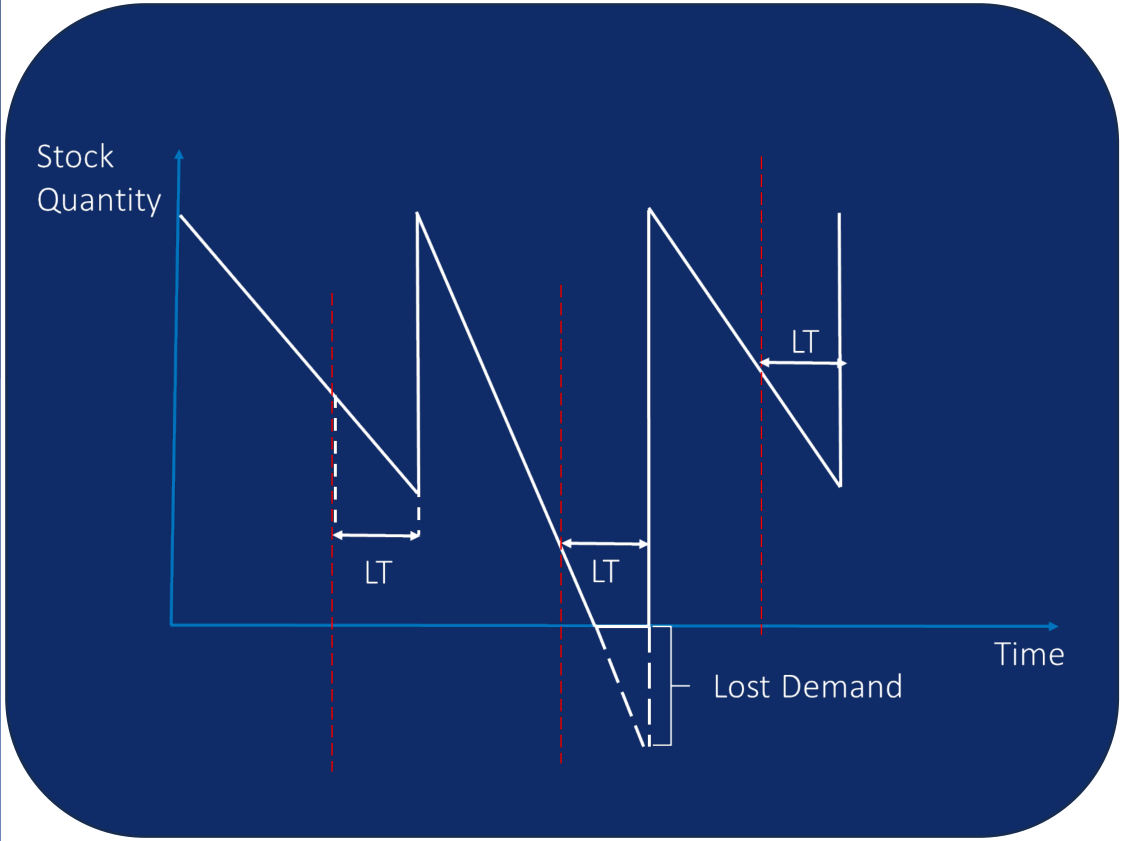


Inventory Management Story
KPIs of inventory Management

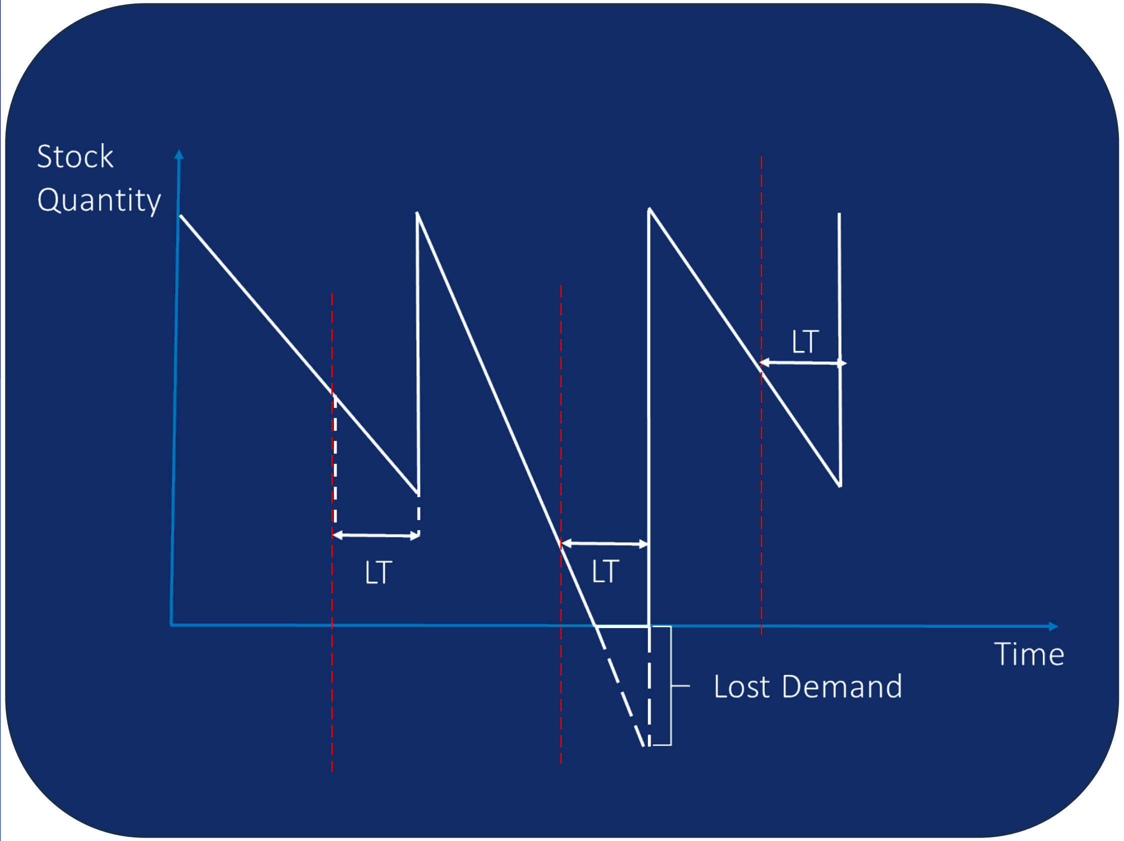




Stock Cost = Stock Quantity * cost_price*coef

Coef = 15%-25%:

Inventory Management Story
KPIs of inventory Management

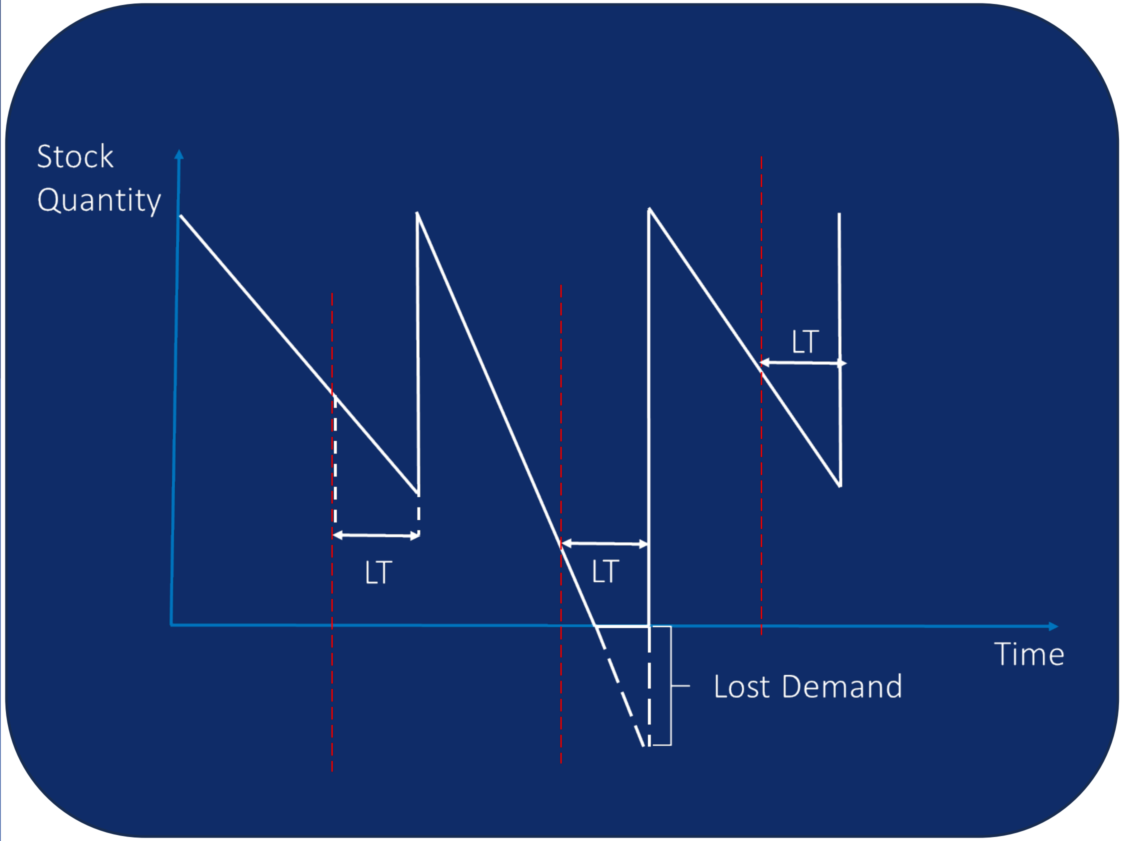




Stock Cost = Stock Quantity * cost_price*coef


Lost Demand Cost = Lost Quantity * price* Margin

Inventory Management Story
KPIs of inventory Management





Stock Cost = Stock Quantity * cost_price*coef


Lost Demand Cost = Lost Quantity * price* Margin



Order
Shipment Cost = Orders Amount * price_shipment

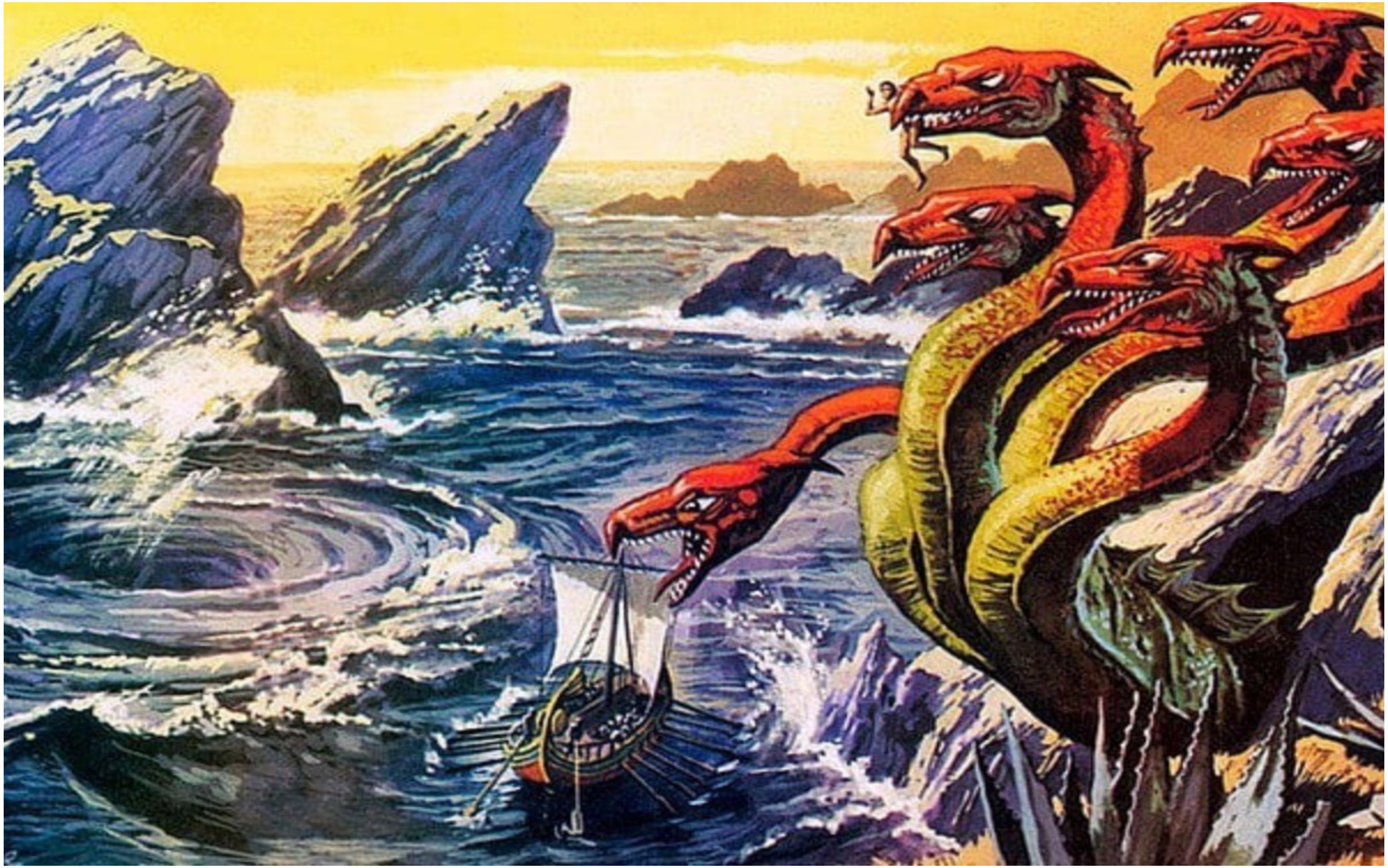
Inventory Management Story
Key KPI of inventory Management

Inventory turnover (days) for the ***T*** previous periods:

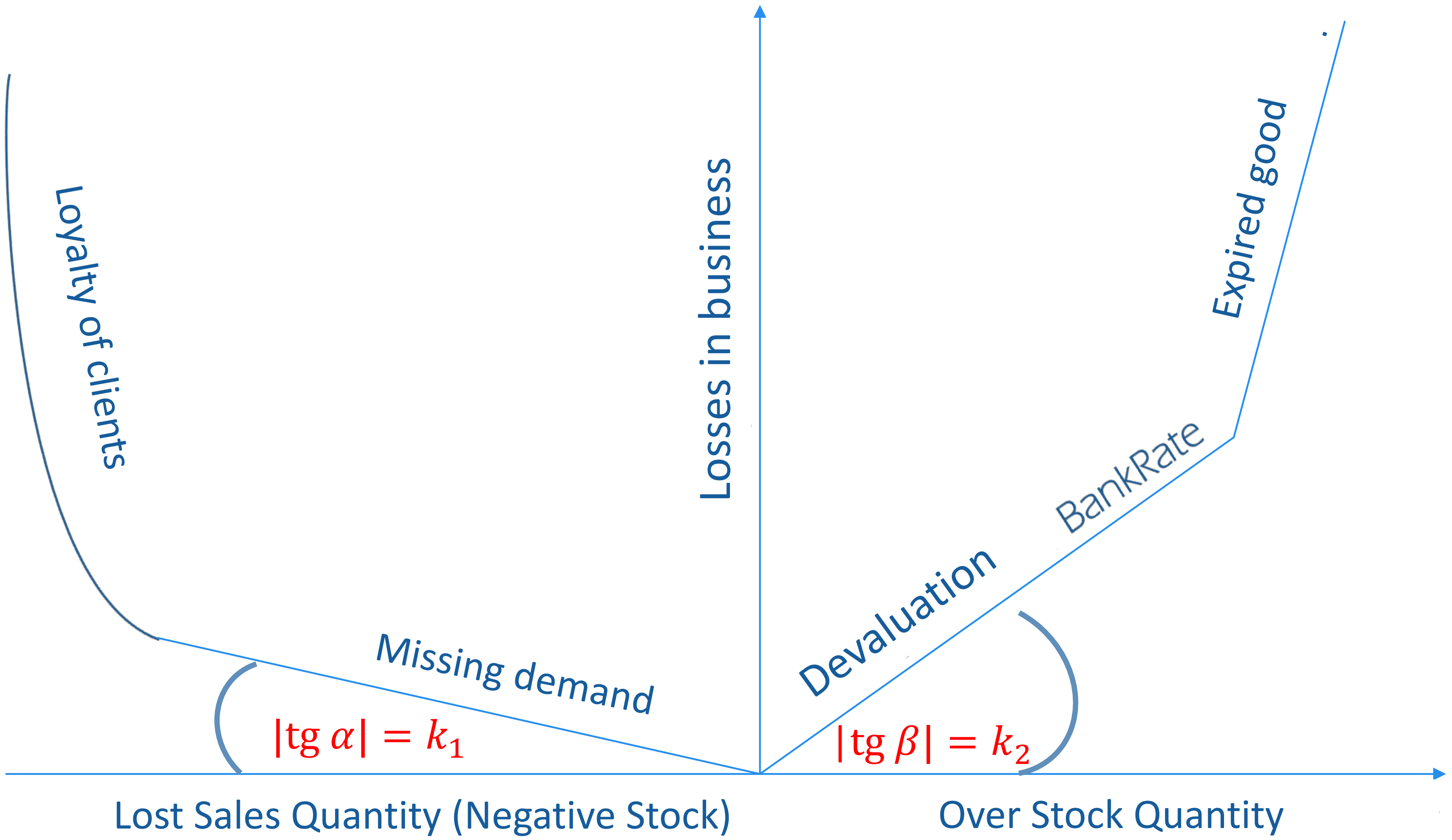
$$\frac{(Inventory\ T\ periods\ back + current\ Inventory)\ /\ 2}{(Sales\ sum\ for\ T\ periods)\ /\ T} =$$

$$= \frac{(Average\ inventory\ per\ 1\ period)}{(Average\ sales\ per\ 1\ period)}$$

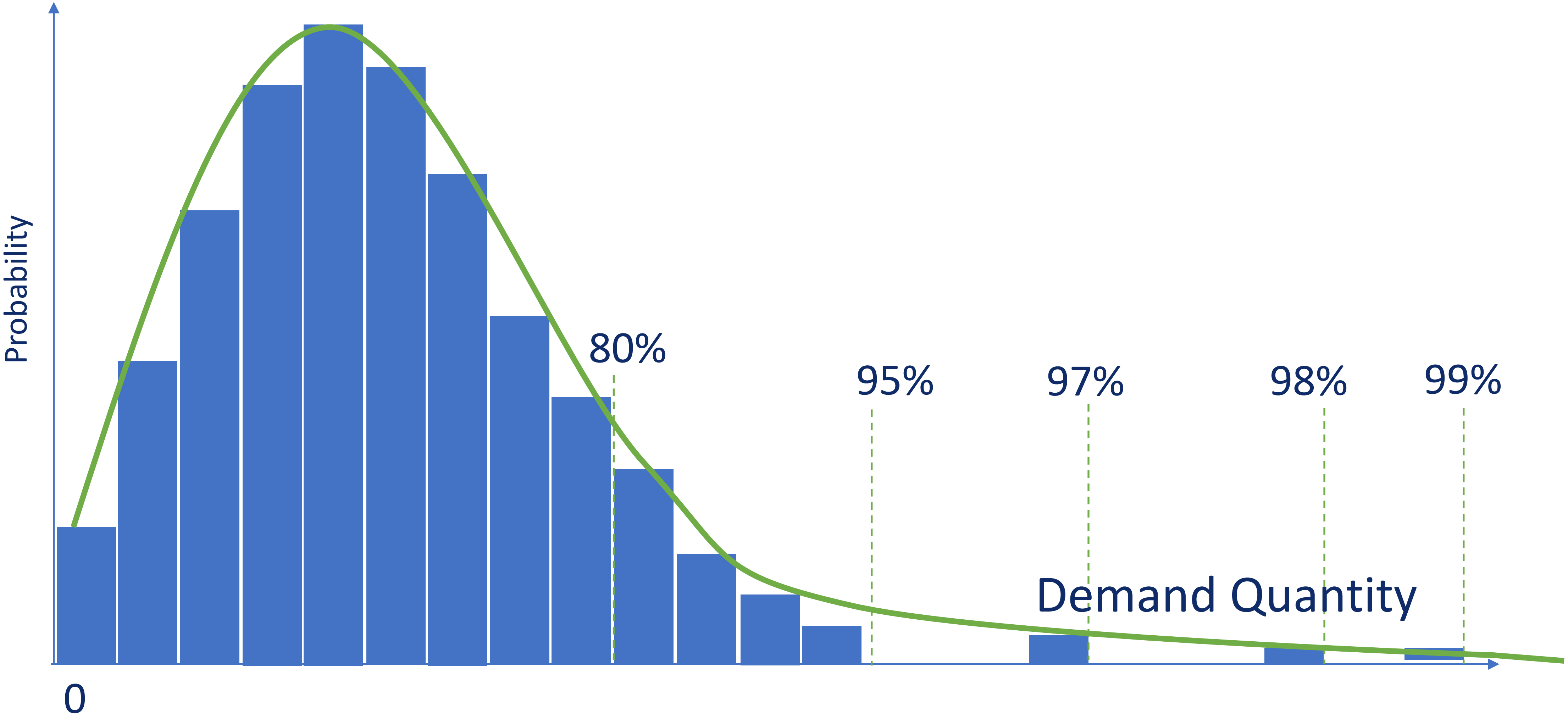
Stock Cost
VS
Lost Demand
Cost Trade-off



Stock Cost VS Lost Demand Cost Trade-off



Inventory Optimization Approach. Service Level



How SL is connected to Stock Cost and Lost Demand Cost?

$$L(y, \hat{y}) = \begin{cases} k_1 |y - \hat{y}|, y \geq \hat{y}; \\ k_2 |y - \hat{y}|, y < \hat{y} \end{cases}$$

y – real demand, $\hat{y}$ - real stock level

$p(y)$ - demand probability distribution

- Optimal Stock Level:

$$\hat{y}^* = \operatorname*{argmin}_{\hat{y}} \int L(y, \hat{y}) \cdot p(y)\, dy$$

$$\hat{y}^* = \Phi^{-1}\left(\frac{k_1}{k_1 + k_2}\right)$$

How SL is connected to Stock Cost and Lost Demand Cost?

$$L(y,\hat{y}) = \begin{cases} k_1|y-\hat{y}|, y \geq \hat{y}; \\ k_2|y-\hat{y}|, y < \hat{y} \end{cases}$$

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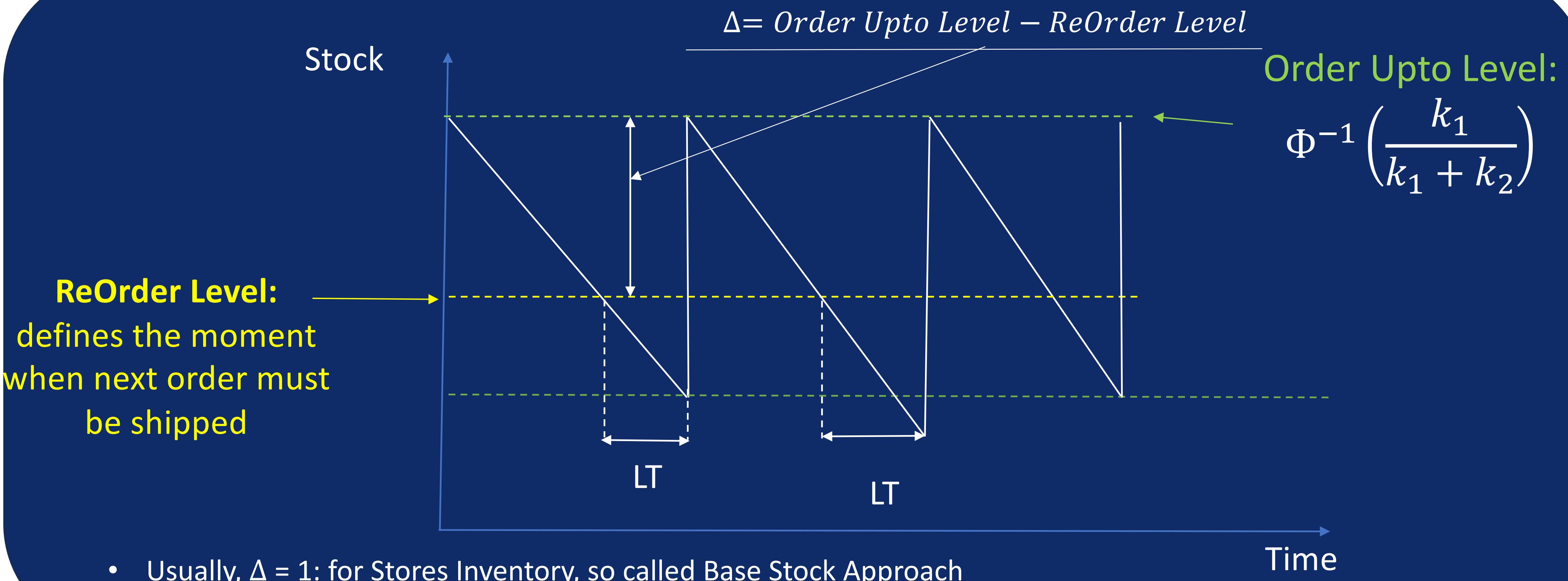
$$\hat{y}^* = \Phi^{-1}\left(\frac{k_1}{k_1+k_2}\right)$$

- Conclusion:

$$\text{Service Level} = \frac{k_1}{k_1+k_2}$$

Inventory Management



- Usually, Δ = 1: for Stores Inventory, so called Base Stock Approach
- Usually, $\Delta = EOQ\ (Wilson\ Formula)$ for Warehouses Inventory, so called Safety Stock Approach

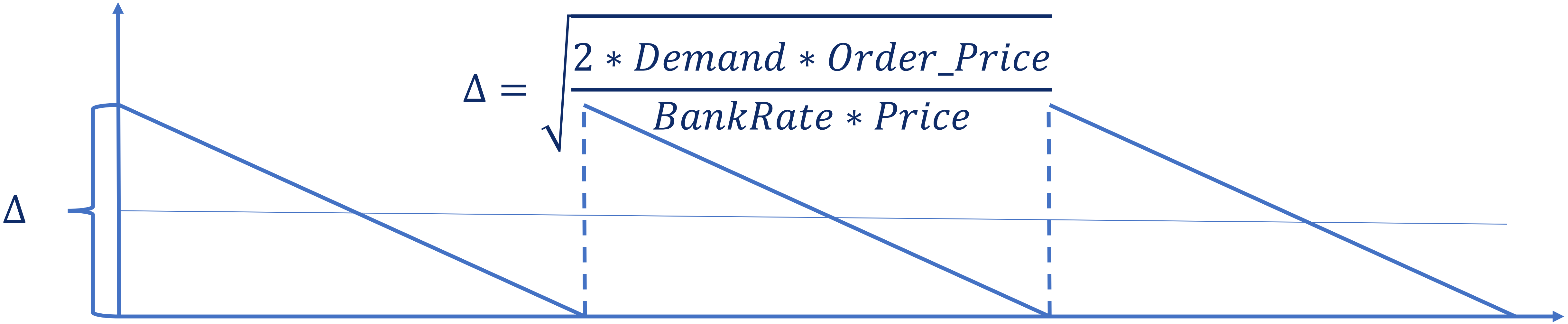
Economic Order Quantity (EOQ)

$$Cost = Buying\_Cost + Order\_Processing\_Cost + Stock\_Holding\_Cost$$

$$Cost = Demand * Price + \frac{Demand}{\Delta} * Order\_Price + Price * \frac{\Delta}{2} * BankRate$$

$$\frac{\delta Cost}{\delta \Delta} = -\frac{Demand * Order\_Price}{\Delta^2} + Price * \frac{1}{2} * BankRate = 0$$

$$\Delta = \sqrt{\frac{2 * Demand * Order\_Price}{BankRate * Price}}$$



Demand=0,035*365;
Price=150,65,
Order_Price = 10,
BankRate = 23,6%

Economic Order Quantity (EOQ)
Real case

$$\Delta = \sqrt{\frac{2 * Demand * Order\_Price}{BankRate * Price}} \qquad \Delta = \sqrt{\frac{2 * 0{,}035 * 365 * 10}{23{,}6\% * 150{,}65}} = 2{,}68 \sim 3$$

$$Cost = Demand * Price + \frac{Demand * Order\_Price}{\Delta} + BankRate * Price * \frac{\Delta}{2}$$

Policy 1: 0-1 ($\Delta = 1$)

$$Cost = \frac{0{,}035 * 365 * 10}{1} + 0{,}236 * 150{,}65 * \frac{1}{2} = 127{,}75 + 17{,}77 = 145{,}5$$

Policy 2: 0-3 ($\Delta = 3$)

$$Cost = \frac{0{,}035 * 365 * 10}{3} + 0{,}236 * 150{,}65 * \frac{3}{2} = 42{,}58 + 53{,}31 = 95{,}89$$

Order Allocation in case of Shortage in Warehouse
Business problem

- SKU stock deficit: there is less stock than shops demand
- There is no priority among shops
- We need to minimize stock turnover: the time the stock passes in the net
- **Which shops will take available stock?**

Stock Distribution in Deficit Conditions
Business-task example

Moscow

Ekaterinburg

Khabarovsk

Stock Distribution in Deficit Conditions
Business-task example

Total = **Desired quantity**

70

=

Moscow

50

+

Ekaterinburg

15

+

Khabarovsk

5

But stock outstanding amounts to 50

Stock Distribution in Deficit Conditions
Business-task example

Total

~~70~~

50

=

Moscow

~~50~~

~~?~~

+

Ekaterinburg

~~15~~

?

+

Khabarovsk

~~5~~

?

Order Allocation in case of Shortage in Warehouse
For each SKU:

Symbol	Business meaning
n	Store quantity within a considered SKU
i	Store ID
$FINAL\_IOW\_ORDER_i$	Desired quantity for store i
INV_i	Current stock and pipelines in store i
$WINV$	Stock in Warehouse for SKU
v_i	Total Demand of SKU in store i
$DEMAND_i(t)$	Demand of SKU in store i within day t
LT_i	Lead time: number of days to deliver stock from warehouse to store i
k_i	Order quant
x_i	Final order to store i

Order Allocation in case of Shortage in Warehouse
Math problem

Optimized function (for SKU):

$$\underbrace{\max_i \left[LT_i + \frac{\max\left(0, INV_i - \sum_{t=1}^{LT_i} Demand_t\right) + k_i \cdot x_i}{v_i} \right]}_{maximal\ turnover\ among\ all\ stores} \longrightarrow \min_{x_i}$$

Business constrains:

- $x_i \in \mathbb{N}\ (x \geq 0$ и $x_i - integer)$
- $x_i \leq FINAL\_IOW\_ORDER_i$
- $\sum_{i=1}^{n} k_i \cdot x_i \cong WINV$

Symbol	Business meaning
n	Store quantity within a considered SKU
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INV_i	Current stock and pipelines in store i
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LT_i	Lead time: number of days to deliver stock from warehouse to store i
k_i	Order quant
x_i	Final order to store i

- Price optimization problem in Retail usually includes several goals (multi-dimensional KPI)
- The key analytical task when approaching prices is the assessment of elasticity, which is solved by building the price-elasticity-demand model.
- The correct solution of the pricing problem essentially depends on business constraints

- The classical approach to inventory optimization problem based on several optimization subproblems.
- Global task to define inventory policy (no limits): order to level, reorder level, safety stock.
- Calculation of optimal orders considers restrictions (stock balances, store priorities, etc.)

Self-check questions

- Indicate two key KPIs for Distribution Logistics
- Explain the business implications of the Order-upto-level and Reorder-level values.
- What is the optimal order size and how is it connected with Order-upto-level and Reorder-level?
- What is the price elasticity of demand?
- List at least 5 different causal variables that can be used to build an elasticity model. For each of them, give at least one example of a problem when their absence in the model will lead to an incorrect definition of elasticity.

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